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Economic Evaluation of Digital Twin Investment in Additive Manufacturing SMEs: An Integrated Real-Options and Multi-Objective Portfolio Approach

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Abstract: Digital twin investment presents a difficult economic decision for small and medium-sized manufacturing enterprises (SMEs), where limited capital must be allocated between production technologies and competing innovation projects. Conventional investment appraisal focuses mainly on direct operational savings and often overlooks how digital twins affect the costs, success probabilities, and risks of other projects. This study investigates the economic value of digital twin adoption within an innovation portfolio and examines whether portfolio-based appraisal leads to different investment decisions from stand-alone evaluation. A Twin-Enabled Innovation Portfolio (TEIP) model was developed by integrating staged real options with multi-objective portfolio optimization. The model accounted for project interdependencies, correlated failures, development costs, and uncertainty in digital twin performance. Non-dominated Sorting Genetic Algorithm II (NSGA-II) was used to optimize expected net present value (NPV) and 5% conditional value-at-risk (CVaR), with investment strategies evaluated through Monte Carlo simulations. Across 30 simulated additive manufacturing SMEs, the integrated approach increased Pareto-frontier hypervolume by 26.1% relative to portfolios without digital twins and by 6.2% relative to stand-alone appraisal. Portfolio re-optimization contributed a further 1.7% compared with portfolio-first sequential evaluation. Approximately 62% of the modelled improvement was associated with effects on development projects rather than direct operational savings. In a separate analysis of 400 simulated firms, stand-alone and portfolio-based appraisals produced different adoption decisions in 24% of cases. Additional confirmation and hold-out tests supported the main findings. The results indicate that evaluating digital twins solely through operational savings can lead to inappropriate technology investment decisions. The proposed framework provides a basis for assessing digital twin investments alongside competing development projects and for identifying adoption thresholds under capital and risk constraints.

Keywords: Digital twin investment; Technology economic evaluation; Additive manufacturing; Real options; Innovation portfolio optimization; Multi-objective optimization; Conditional value-at-risk; Technology adoption

JEL Classification: O32, G31, C61, L23, O33

1. Introduction

Digital twins have become an important component of digital manufacturing, connecting physical production processes with computational models for monitoring, analysis, and operational decision-making (Kritzinger et al., 2018; Tao et al., 2019). Their adoption, however, requires substantial investment in data acquisition, modelling, system integration, and maintenance. For small and medium-sized manufacturing enterprises (SMEs), these expenditures compete with product development, process improvement, and other innovation activities for limited financial resources. The economic justification for a digital twin therefore depends not only on its operational performance but also on how the investment affects the allocation and expected returns of the firm's development portfolio.

This issue is particularly relevant to additive manufacturing, where production quality, material consumption,

and process stability are closely connected. Defective builds and repeated qualification trials can impose considerable costs, while changes in process parameters may affect the commercial prospects of several development projects. A digital twin can contribute to production efficiency through process monitoring and defect reduction, but its economic role may extend beyond these direct savings. Information obtained from the twin can also support process qualification, reduce development expenditure, and improve the probability that projects progress through successive development stages. These effects connect the value of the technology to the performance of the projects that use it.

Conventional capital investment appraisal generally evaluates a technology through the discounted value of its expected cash flows. Real-options analysis extends this approach by accounting for managerial flexibility, including the ability to defer, stage, or expand an investment as information becomes available (Dixit & Pindyck, 1994; Trigeorgis, 1996). For digital twins, staged deployment is especially relevant because firms can begin with a limited implementation and subsequently decide whether a more extensive system is economically justified. Nevertheless, evaluating this investment separately from the innovation portfolio leaves an important part of the decision outside the appraisal. The digital twin may change the expected returns, development costs, and risk characteristics of other projects, while the composition of the portfolio determines how much value the twin can generate.

The relationship is therefore reciprocal. A firm with several process-related development projects may obtain substantial portfolio-level value from a digital twin even when its direct operational savings are insufficient to justify the initial expenditure. Conversely, adopting the technology can consume capital that would otherwise finance projects with higher expected returns. These competing effects make the adoption decision inseparable from project selection when the investment budget is constrained. They also raise a question about the appropriate sequence of decisions: whether the twin should be evaluated independently, assessed against an existing portfolio, or selected jointly with the projects it supports.

This study addresses these issues through an integrated economic evaluation of digital twin investment in additive manufacturing SMEs. It develops a Twin-Enabled Innovation Portfolio (TEIP) model that combines staged real options with multi-objective portfolio optimization. The model represents the effects of digital twin adoption on direct operating savings, project success probabilities, development costs, and dependence between project outcomes. It also incorporates a staged upgrade decision based on an observed efficacy signal. Expected net present value (NPV) and conditional value-at-risk (CVaR) are used to examine the economic trade-off between portfolio returns and downside exposure. By comparing independent, sequential, and integrated investment strategies, the analysis identifies how digital twin valuation changes when its effects on other development projects are included.

1.1 Related Work and Research Gap

Research relevant to digital twin investment appraisal spans manufacturing digitalization, real-options theory, and innovation portfolio management. These fields provide complementary foundations for evaluating technology investments, although they approach the relationship between technological capabilities and project-level economic performance from different perspectives.

Digital twin research has established the technological foundations for linking physical systems with digital representations. Grieves & Vickers (2017) discussed the role of digital twins in managing complex system behaviors, while Kritzinger et al. (2018) classified digital twin applications in manufacturing. Tao et al. (2019) examined the development of digital twins in industrial systems, and Fuller et al. (2020) reviewed their enabling technologies and implementation challenges. In additive manufacturing, Kantaros et al. (2022) discussed digital twin applications and the technical conditions affecting their implementation. Together, these studies establish the relevance of digital twins to manufacturing operations and provide a basis for examining their economic effects.

The investment decision introduces a different set of considerations. Manufacturing firms must compare the expected operational gains from a digital twin with the expenditure required for implementation and continued operation. Thomas (2024) presented an economic assessment of digital twins based on their costs and expected returns, providing a basis for technology-level appraisal. For SMEs, Yasin et al. (2021) proposed a roadmap for digital twin integration that addressed the practical requirements of adoption. These studies connect digital twin implementation with economic and organisational decisions. Their principal focus, however, remains the technology investment or its adoption process. A digital twin that changes the performance of several development projects requires an additional level of appraisal, particularly when those projects compete for the same financial resources.

Real-options theory provides a foundation for incorporating investment timing and managerial flexibility into this evaluation. Dixit & Pindyck (1994) and Trigeorgis (1996) established the treatment of irreversible investment decisions under changing information and uncertain returns. Bowman & Hurry (1993) extended the options perspective to strategic resource investment, while Kogut & Kulatilaka (2001) examined organisational capabilities as sources of option value. Trigeorgis & Reuer (2017) subsequently connected real-options reasoning

with strategic management decisions. More recently, Savolainen & Knudsen (2025) considered real options alongside strategic foresight in the transition towards Industry 5.0. This literature establishes why staged investment and learning can affect technology valuation. For digital twin adoption, it supports a decision structure in which an initial implementation is followed by an upgrade decision informed by observed performance. The economic value of that flexibility must also be considered alongside the projects whose returns depend on the technology.

Innovation portfolio research provides the corresponding framework for allocating capital among competing projects. Markowitz (1952) established the relationship between expected returns and portfolio risk, while Cooper et al. (1999) examined portfolio management practices in new product development. Ghasemzadeh & Archer (2000) developed a decision-support approach to project portfolio selection, and Rockafellar & Uryasev (2000) introduced CVaR optimization as a means of addressing adverse outcomes. The importance of financing constraints for innovation investment was further examined by Hall & Lerner (2010). More recently, Kjærgaard-Nielsen et al. (2024) presented a multi-objective approach to industrial research and development portfolio selection that considered project characteristics and interdependencies. These contributions provide the basis for selecting projects under financial and risk constraints. In the present investment setting, the portfolio structure must additionally account for a technological asset that changes the economic characteristics of the projects available for selection.

The connection between these research streams concerns the role assigned to the digital twin. In technology-level appraisal, its value is primarily associated with its own operating costs and savings. In real-options analysis, the investment decision incorporates flexibility and learning. In portfolio optimizations, projects are selected according to their expected returns, resource requirements, and risk characteristics. Combining these perspectives allows the digital twin to be evaluated as an enabling investment whose adoption changes the performance of the portfolio itself. The relevant economic comparison is then between portfolios constructed under alternative technology decisions, rather than between the direct savings and costs of the digital twin alone.

This distinction also affects the interpretation of risk. A digital twin may reduce exposure to common sources of project failure, but the economic consequences depend on relationships between projects. When successful projects generate complementary returns, changes in their dependence structure can affect the expected value of those complementarities. Portfolio-level appraisal must therefore consider both downside risk and the joint occurrence of successful project outcomes. Similarly, the value of a staged upgrade depends on how the resulting technological capabilities influence the projects funded by the firm, rather than solely on the additional operating savings generated by the upgraded system.

Table 1. Positioning of the study within the relevant literature

Research Area	Representative Studies	Main Contribution to Investment Appraisal	Focus of the Present Study
Digital twins in manufacturing	Fuller et al. (2020); Grieves & Vickers (2017); Kantaros et al. (2022); Kritzinger et al. (2018); Tao et al. (2019)	Digital twin architectures, industrial applications, and manufacturing process integration	Economic effects of digital twin adoption on production and development projects
Technology adoption and economic evaluation	Thomas (2024); Yasin et al. (2021)	Implementation requirements, adoption planning, and technology-level cost–benefit assessment	Investment appraisal under competing capital requirements
Real options and strategic investment	Bowman & Hurry (1993); Dixit & Pindyck (1994); Kogut & Kulatilaka (2001); Savolainen & Knudsen (2025); Trigeorgis (1996); Trigeorgis & Reuer (2017)	Investment flexibility, staged commitment, learning, and capability valuation	Staged digital twin deployment and upgrade decisions within an innovation portfolio
Innovation portfolio optimization	Cooper et al. (1999); Ghasemzadeh & Archer (2000); Hall & Lerner (2010); Kjærgaard-Nielsen et al. (2024); Markowitz (1952); Rockafellar & Uryasev (2000)	Project selection, capital allocation, interdependencies, and risk management	Joint selection of enabling technology and development projects under NPV–CVaR objectives
Present study	TEIP model	Integrated real-options and multi-objective economic evaluation	Portfolio-level technology valuation, investment strategy comparison, adoption thresholds, and sources of investment value

Note: TEIP = Twin-Enabled Innovation Portfolio; NPV = net present value; CVaR = conditional value-at-risk.

The resulting research gap lies in the integrated economic treatment of these decisions. The established literature provides methods for digital twin appraisal, staged investment, and risk-constrained portfolio selection, but their combination is needed to quantify the economic effects of an enabling technology across a development portfolio.

In particular, the investment framework must distinguish the value obtained by evaluating the twin against an existing optimized portfolio from the additional value obtained by jointly reconsidering technology adoption and project selection. It must also identify the contribution of individual economic channels, the conditions governing adoption, and the consequences of using stand-alone appraisal across firms with different production and development characteristics. Table 1 positions the present study against these research foundations.

1.2 Research Objectives and Contributions

The objective of this study is to establish an economic evaluation framework for digital twin investment that accounts for its direct operating returns and its effects on the wider innovation portfolio. The analysis examines how investment decisions change when the twin is treated as an enabling technological asset rather than an independent capital project. It also investigates how staged deployment, project interdependencies, capital constraints, and downside risk influence the economic case for adoption.

The study makes four contributions to the evaluation of manufacturing technology investment.

First, it develops an integrated valuation model in which digital twin adoption and innovation project selection are linked through their economic consequences. The TEIP model incorporates a staged technology investment, project-level success probabilities, development expenditure, correlated project outcomes, and complementary returns. The investment decision is evaluated against expected NPV and CVaR, allowing changes in the portfolio's risk–return characteristics to be examined alongside the direct savings generated by the twin.

Second, the study distinguishes the economic consequences of alternative appraisal procedures. By comparing portfolios without digital twins, stand-alone technology appraisal, portfolio-first sequential evaluation, and joint optimizations, it separates the value of recognizing portfolio-level effects from the additional value of reselecting projects after the technology decision. Channel ablation and sensitivity analysis identify the relative roles of operational savings, project success, development costs, failure dependence, and staged upgrading. This provides a more detailed account of technology investment value than an aggregate NPV comparison.

Third, the study derives analytical relationships that support practical investment screening. A closed-form break-even condition links the stand-alone investment decision to recoverable scrap value and the cost of digital twin ownership. A corresponding portfolio-aware threshold incorporates the economic contribution of development projects. The analysis also examines how dependence between project outcomes affects complementary returns and how information about technology efficacy changes the value of a staged upgrade. These relationships connect the optimization results to interpretable investment conditions.

Fourth, the framework is examined across heterogeneous simulated manufacturing firms and alternative economic settings. Population-level comparisons identify differences between stand-alone and portfolio-aware adoption decisions, while confirmation samples, hold-out evaluations, and robustness tests assess the behaviors of the model under different firm characteristics. The analysis also considers downside exposure and the consequences of investment subsidies and research tax credits. These experiments extend the valuation framework from individual portfolio decisions to technology adoption patterns and capital allocation outcomes.

The contribution of the study is thus an integrated approach to technology investment appraisal in which the value of a digital twin depends on both its own cash flows and the development portfolio it supports. The framework provides a basis for comparing investment strategies, identifying economically justified adoption conditions, and allocating limited capital between manufacturing technology and innovation projects.

The remainder of the paper is organized as follows. Section 2 presents the digital twin investment and portfolio valuation model, including its analytical properties. Section 3 describes the optimization procedure, benchmark strategies, simulation design, and validation methods. Section 4 reports the economic evaluation results, investment thresholds, portfolio effects, learning value, and policy analysis. Section 5 discusses the implications for technology investment appraisal and manufacturing management, together with the limitations and directions for further research. Section 6 concludes the study.

2. Digital Twin Investment and Portfolio Valuation Model

The TEIP model evaluates digital twin investment as part of a firm's capital allocation decision. It combines the direct operating returns of the technology with its effects on innovation projects, including development expenditure, stage-gate success, and dependence between project outcomes. The model also allows the firm to deploy a limited digital twin initially and upgrade it after receiving information about its performance. Investment strategies are compared in terms of expected NPV and downside risk under a common budget constraint.

The framework consists of a project portfolio, a stochastic representation of project outcomes, a staged digital twin investment, and a multi-objective valuation procedure. Figure 1 illustrates the relationships between these components. Time is measured in years, monetary quantities are expressed in USD million unless otherwise stated, and the principal parameters and symbols are listed in Appendix.

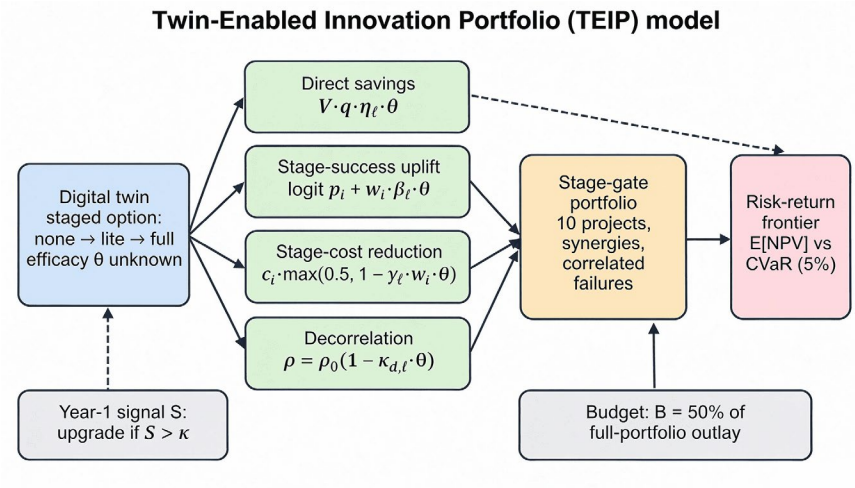


Figure 1. Structure of the Twin-Enabled Innovation Portfolio framework

Note: TEIP = Twin-Enabled Innovation Portfolio; NPV = net present value; CVaR = conditional value-at-risk. The subscript ℓ denotes the digital twin implementation level (lite or full). θ represents technology efficacy. For the four economic channels: V and q denote annual production value and defect rate, respectively; η_ℓ, β_ℓ, and γ_ℓ are the savings, success-uplift, and cost-reduction coefficients at level ℓ; p_i, c_i, and w_i represent the baseline success probability, stage cost, and effective twin exposure for project i; ρ and ρ_0 are the adjusted and baseline correlations between project outcomes, and $\kappa_{d,l}$ is the decorrelation coefficient. S and κ denote the year-1 signal and the upgrade threshold, respectively. B represents the available investment budget.

2.1 Firm Characteristics and Innovation Portfolio

Consider an additive manufacturing SME with $N = 10$ candidate innovation projects. Each project represents a product or process development activity requiring a sequence of technical and commercial decisions before generating revenue. Project i has $n_i \in \{2, 3, 4\}$ remaining development stages, a cost c_i per stage, a baseline stage-success probability p_i, and annual revenue R_i over $L = 5$ years following commercialization. A project proceeds to the next stage only after completing the preceding one. Failure terminates further development expenditure, introducing an abandonment option into the investment process.

The projects differ in their exposure to digital twin technology. Process-related projects can use the twin directly for production monitoring, parameter adjustment, and process qualification. Other projects may receive indirect support through improved manufacturing information and reduced design iterations. A binary indicator ϕ_i distinguishes these project types, with $\phi_i = 1$ for process-related projects and $\phi_i = 0$ otherwise. The effective exposure of project i is defined as

$$w_i = \phi_i + 0.3(1 - \phi_i) \tag{1}$$

Accordingly, process-related projects receive the full modelled effect of the digital twin, while other projects receive 30% of that effect. This distinction allows the economic contribution of the technology to depend on the composition of the firm's development portfolio.

Project returns may also be interdependent. Eight project pairs (i, j) are selected to represent complementary development activities. For each pair, the parameter s_{ij} specifies an additional annual return of s_{ij} min (R_i, R_j) when both projects complete their development stages successfully. These returns represent commercial opportunities arising from the joint availability of related products or manufacturing capabilities. They are included in the portfolio cash flows only when the corresponding projects succeed.

For example, a metal additive manufacturing firm may undertake a process-monitoring project alongside the development of a lattice-structured heat exchanger. The monitoring project may involve sensor trials, pilot builds, and process qualification, while the product project may require design verification and customer approval. Digital twin adoption can reduce development costs in both activities, although its direct influence is greater for process monitoring. Successful completion of the two projects can also generate complementary revenue if the qualified manufacturing process supports commercial production of the new component. The economic value of the digital twin therefore depends on the projects selected and, on the returns, generated by their interactions.

This portfolio representation links technology adoption to two distinct economic effects. The first is a change in the expected performance of individual projects. The second is a change in portfolio value arising from the joint outcomes of complementary projects. Both effects enter the investment evaluation alongside the direct operating savings of the digital twin.

2.2 Correlated Project Outcomes and Interdependencies

Innovation projects face technical and commercial risks that may be shared across development activities. Changes in manufacturing conditions, common technical requirements, and broader operating circumstances can affect several projects simultaneously. The TEIP model represents this dependence through a one-factor Gaussian structure, following the approach used in correlated portfolio modelling (Vasicek, 2002).

Stage s of project i succeeds when the latent outcome variable u_{is} is below the threshold associated with its effective success probability:

$$u_{is} = \sqrt{\rho}\, Z + \sqrt{1-\rho}\, \varepsilon_{is} < \Phi^{-1}\left(p_i^{\text{eff}}\right) \tag{2}$$

Here $Z \sim N(0,1)$ is a systematic factor shared by all project stages, $\varepsilon_{is} \sim N(0,1)$ is an independent stage-specific factor, and Φ^{-1} denotes the inverse standard normal cumulative distribution function. The parameter p_i^{eff} is the stage-success probability after accounting for the digital twin's effect. Because the latent variable has a standard normal marginal distribution, the probability of completing an individual stage is p_i^{eff}.

The parameter ρ controls dependence between stage outcomes. In the absence of a digital twin, it equals the baseline value ρ_0. Following adoption, the technology can reduce this dependence through its decorrelation channel. The adjusted parameter is

$$\rho = \min\left\{\rho_0, \max\left[0.1\rho_0, \rho_0\left(1 - \kappa_{\mathrm{d},\ell}\theta\right)\right]\right\} \tag{3}$$

The coefficient κ_d, ℓ denotes the decorrelation effect at digital twin level ℓ, and θ represents realised technology efficacy. The lower bound prevents dependence from falling below 10% of its baseline value; the upper bound prevents the intervention from increasing it.

The common factor introduces dependence both between projects and between successive stages of the same project. Consequently, the probability that a project completes all its stages depends on the joint distribution of stage outcomes, not simply on the product of their marginal success probabilities. This distinction matters when valuing project complementarities, since additional revenue is realised only when both members of a complementary pair complete their development activities.

Commercial revenue is subject to a separate stochastic component. The model combines a common lognormal revenue shock with dispersion parameter 0.30 and a project-specific lognormal shock with dispersion parameter 0.25. The former represents market conditions affecting multiple projects, while the latter captures differences in their commercial outcomes. Together with correlated stage completion, these shocks generate variation in portfolio NPV and its lower tail.

Reducing dependence between development outcomes can change both the distribution of portfolio losses and the probability of joint project success. The net economic effect is therefore evaluated through the portfolio objectives rather than assigned a positive value in advance.

2.3 Digital Twin Investment and Staged Upgrade Options

The digital twin investment is represented by three operating states: no twin, lite twin, and full twin. The lite system provides an initial level of process monitoring and decision support, while the full system represents a more extensive implementation with greater operating and project-level effects.

The lite twin requires initial capital expenditure of USD 0.25 million and annual operating expenditure of USD 0.03 million. The corresponding costs of a full twin are USD 0.80 million and USD 0.10 million per year. A firm that initially adopts the lite system can upgrade to the full system at the end of year 1 by paying an additional USD 0.65 million. The investment decision therefore includes both the initial commitment and the subsequent choice of whether to expand technological capability.

Realised efficacy θ follows a lognormal distribution with mean 1 and a coefficient of variation (CV) ranging from 0.3 to 0.8 across firms. The parameter scales the modelled effects of the technology, allowing operating and development outcomes to differ across scenarios. At the end of the first year, the firm receives a noisy signal of efficacy:

$$S = \theta \exp\left(0.35\varepsilon_S - \frac{0.35^2}{2}\right) \tag{4}$$

where, $\varepsilon_s \sim N(0, 1)$ is the independent signal-noise term. The firm upgrades when $S > \kappa$, where κ is the selected upgrade threshold.

The digital twin affects the firm through four economic channels. Direct savings arise from reductions in recoverable production losses. The remaining channels operate through the development portfolio by changing

stage-success probabilities, stage costs, and dependence between project outcomes. Their functional forms and parameters are given in Table 2.

The direct-savings channel links digital twin adoption to recoverable scrap value, Vq. The remaining channels connect adoption to project selection. A higher stage-success probability changes expected commercial returns, while lower stage expenditure changes expected development costs and resource requirements. The decorrelation channel alters joint project outcomes and therefore affects both downside exposure and complementary returns.

Table 2. Economic channels and parameters of digital twin investment

Channel	Mechanism	Lite	Full	Basis
Direct savings	$Vq\eta_\ell\theta$ per year over $H = 8$ years	$\eta_\ell = 0.20$	$\eta_\ell = 0.45$	Modelled reduction in production losses; mechanism supported by Kantaros et al. (2022).
Success uplift	$\text{logit}(p_i^{\text{eff}}) = \text{logit}(p_i) + w_i\beta_\ell\theta$	$\beta_\ell = 0.25$	$\beta_\ell = 0.55$	Stage-gate representation based on Cooper (1990); intervention coefficients specified as modelling assumptions.
Cost reduction	$c_i^{\text{eff}} = c_i \max(0.5, 1 - \gamma_\ell w_i\theta)$	$\gamma_\ell = 0.08$	$\gamma_\ell = 0.18$	Modelled reduction in development-stage expenditure.
Decorrelation	ρ determined by Eq. (3)	$\kappa_{d,\ell} = 0.15$	$\kappa_{d,\ell} = 0.35$	Modelled reduction in dependence between project-stage outcomes.

Note: The intervention coefficients are modelling assumptions rather than empirical estimates. The cited studies provide the conceptual basis for the corresponding mechanisms but do not establish their numerical magnitudes. The effects of these parameters are examined through channel-ablation, sensitivity, and robustness analyses.

These mechanisms operate at different stages of the investment. The lite twin influences the portfolio after initial deployment. If the efficacy signal exceeds the upgrade threshold, full-level parameters apply following the year-1 upgrade. The upgrade decision thus affects future operating savings and development activities that remain exposed to the technology.

The staged structure allows the firm to compare the cost of initial commitment with the value of obtaining information before expanding the investment. Its economic contribution is evaluated jointly with the portfolio rather than through direct operating savings alone.

2.4 Economic Objectives and Budget Constraints

The investment decision consists of project selection, initial digital twin adoption, and the upgrade policy. Let $x = (x_1, \ldots, x_n)$, $x_i \in \{0, 1\}$, denote project selection, with $x_i = 1$ for funded projects. The binary variable $d_0 \in \{0, 1\}$ indicates whether the lite twin is deployed at the start. Conditional on deployment, $\kappa \in [0.2, 2.5]$ determines the upgrade decision at the end of year 1.

For scenario m, portfolio NPV is calculated from discounted project revenues, complementary returns, development expenditure, digital twin operating savings, and ownership costs:

$$\text{NPV}_m = \sum_{t=0}^{T} \frac{\text{CF}_{m,t}^{\text{proj}} + \text{CF}_{m,t}^{\text{syn}} - C_{m,t}^{\text{stage}} + G_{m,t}^{\text{twin}} - K_{m,t}^{\text{twin}}}{(1+r)^t} \tag{5}$$

The cash-flow terms represent project revenue, complementary project revenue, stage expenditure, direct operating savings, and twin capital and operating expenditure, respectively. The discount rate is $r = 10\%$, and T denotes the evaluation horizon. Stage expenditure stops when a project fails. Digital twin adoption modifies development costs and success probabilities according to Table 2.

The firm seeks to maximise expected NPV while controlling exposure to adverse outcomes. The two objectives are:

$$\min_{(x,d_0,\kappa)} (f_1, f_2),\ f_1 = -\text{E}[\text{NPV}],\ f_2 = \text{CVaR}_\alpha(-\text{NPV}) \tag{6}$$

The first objective represents expected economic return. The second measures downside risk through conditional value-at-risk (CVaR), using portfolio loss $L = -\text{NPV}$ and tail probability $\alpha = 0.05$. Following Rockafellar & Uryasev (2000), CVaR is defined as:

$$\text{CVaR}_\alpha(L) = \min_{z\in\mathbb{R}} \left\{ z + \frac{1}{\alpha}\mathbb{E}[(L-z)^+] \right\},\ \alpha = 0.05 \tag{7}$$

where, z is an auxiliary loss threshold and $(L - z)^+ = \max(L - z, 0)$. The tail probability was set to $\alpha = 0.05$. The portfolio must also satisfy the budget constraint.

$$\mathrm{E}[\mathrm{PV}^{\mathrm{out}}(x, d_0, \kappa)] \leq B, B = 0.5\mathrm{E}[\mathrm{PV}_{\mathrm{all}}^{\mathrm{out}}] \tag{8}$$

Here $\mathrm{PV}^{\mathrm{out}}$ is the present value of expenditure associated with the selected investment strategy, and $\mathrm{PV}_{\mathrm{all}}{}^{\mathrm{out}}$ is the expenditure required to fund all candidate projects. The budget is set at 50% of expected full-portfolio outlay, imposing a substantial restriction on project selection and digital twin adoption.

The budget links the technology decision to the innovation portfolio. Initial digital twin expenditure reduces resources available for development projects, while changes in project success and stage costs alter the expected financial consequences of the selected portfolio. The resulting trade-off is evaluated over the Pareto frontier of expected NPV and CVaR.

The model uses a common systematic factor and a spill-over weight of 0.3 to represent project dependence and indirect technological effects. Their influence is examined through the sensitivity and robustness analyses in Section 4.

2.5 Analytical Properties of Investment Decisions

The stochastic portfolio model permits analytical results that clarify the economic mechanisms underlying digital twin adoption. The first establishes a stand-alone investment threshold based on direct operating savings. The second translates the difference between stand-alone and portfolio-aware thresholds into an equivalent economic value. The third examines the effect of dependence between project outcomes on complementary returns.

Proposition 1. Stand-alone break-even condition.

Consider a lite digital twin with initial capital cost K, annual operating cost k, and annual direct savings $V \cdot q \cdot \eta \cdot \theta$ over H years. Let the discount rate be r and define the annuity factor $A_H(r) = (1 - (1 + r)^{-H}) / r$. When the upgrade option and portfolio-level effects are excluded, expected stand-alone NPV is positive if and only if:

$$V \cdot q > \tau_{\mathrm{SA}} = \frac{K + kA_H(r)}{\eta \mathrm{E}[\theta] A_H(r)} \tag{9}$$

Proof. Expected NPV = η E[θ] $A_H(r)$ $V \cdot q$-K-k $A_H(r)$, which is linear and increasing in $V \cdot q$.

The break-even condition links adoption to an observable production measure and the cost of ownership. A 10% proportional increase in ownership costs raises the threshold by 10%, while a 10% reduction in expected efficacy raises it by approximately 11.1%. For the reference lite twin, $K = 0.25$, $k = 0.03$, $\eta = 0.20$, $H = 8$, $r = 10\%$, and E[θ] = 1. The annuity factor is 5.335, giving ownership costs of approximately USD 0.410 million and a stand-alone threshold of 0.384 USD million per year. This is the analytical benchmark for Section 4.4.

Corollary 1. Portfolio externality and the investment threshold.

Suppose the twin also creates a portfolio benefit with present value Π beyond its direct savings, that this benefit enters the firm's objective additively and at its expected value, and that direct savings enter at their expected value as in Proposition 1. Then the integrated hurdle is $\tau_{\mathrm{INT}} = \tau_{\mathrm{SA}} - \Pi / (\eta \mathrm{E}[\theta] A_H(r))$, or equivalently $\Pi = (\tau_{\mathrm{SA}} - \tau_{\mathrm{INT}}) \cdot \eta \mathrm{E}[\theta] \cdot A_H(r)$. Proof. The twin is worth adopting when $V \cdot q \cdot \eta \mathrm{E}[\theta] A_H(r) + \Pi > K + k A_H(r)$. Dividing by η E[θ] $A_H(r)$ and using Eq. (9) gives $V \cdot q > \tau_{\mathrm{SA}} - \Pi / (\eta \mathrm{E}[\theta] A_H(r))$, which defines τ_{INT}. For the reference lite twin, $\tau_{\mathrm{SA}} = 0.384$ and $\tau_{\mathrm{INT}} \approx 0.17$ give $\Pi \approx 0.23$ USD m, about 56% of the 0.41 USD m cost of ownership. This identity expresses the gap between the two thresholds in money; the simulated τ_{INT} already contains the portfolio effect it prices. Because expected NPV is linear in θ, the stand-alone hurdle in Proposition 1 does not depend on the efficacy uncertainty CV_θ. That uncertainty matters only through the upgrade option and the tail, which Proposition 1 leaves out.

Proposition 2. Dependence between project outcomes and complementary returns.

Consider projects i and j with fixed marginal stage-success probabilities and non-negative complementary returns. Under Eq. (2), the probability that both complete all development stages is non-decreasing in the common correlation parameter ρ:

$$\frac{\partial}{\partial \rho} \Pr\left(u_k \leq a_k \text{ for all } k \in \mathcal{K}_{ij}\right) \geq 0 \tag{10}$$

Here $\mathcal{K}_{ij}$ is the set of stages belonging to both projects, and $a_k = \Phi^{-1}(p_k^{\mathrm{eff}})$ is the stage-success threshold. Proof. The latent outcomes are jointly Gaussian with unit marginal variances and common non-negative pairwise correlation ρ. For fixed thresholds, increasing this correlation weakly increases the joint lower-orthant probability, consistent with Slepian (1962). Successful completion requires all relevant thresholds to be met, so joint completion probability is non-decreasing in ρ. With non-negative synergy coefficients, expected complementary

returns are also non-decreasing in ρ.

This distinguishes reducing clustered failures from increasing the expected value of complementary investments. With three stages per project, marginal stage-success probability 0.70, and $\rho_0 = 0.35$, both projects complete with probability approximately 0.266. A full twin with $\kappa_d = 0.35$ and $\theta = 1$ reduces ρ to 0.2275 and joint completion probability to approximately 0.214, a reduction of about 20%. Under independent stage outcomes, joint completion probability is approximately 0.118.

The economic consequence depends on the balance between changes in downside exposure and complementary returns. In the modelled portfolios, the reduction in expected synergy value outweighed the associated improvement in tail risk. The channel-ablation analysis in Section 4.2 quantifies this effect.

Together, these results establish the economic basis for portfolio-level appraisal. The stand-alone threshold identifies the direct operating return needed to recover ownership costs, while the portfolio-aware threshold incorporates development-project contributions. The dependence result explains how project interactions alter the value of technological intervention. These relationships underpin the optimization procedure and strategy comparisons in Section 3.

3. Optimization Method and Experimental Design

The economic evaluation was conducted through a simulation-based optimization framework that compared alternative digital twin investment strategies under identical firm characteristics and stochastic conditions. The experimental design combined multi-objective portfolio optimization, independent scenario evaluation, sensitivity analysis, and population-level investment appraisal. This arrangement allowed the economic contribution of digital twin adoption to be distinguished from the effects of project selection, investment sequencing, and optimization procedure.

The main analysis covered 30 heterogeneous additive manufacturing SMEs generated from the parameter ranges in Table A1. Each investment strategy was optimized using 3,000 Monte Carlo scenarios and subsequently evaluated on 20,000 independent scenarios. Expected NPV and CVaR were used to construct the risk–return frontiers, while hypervolume and expected NPV at matched risk levels provided the principal measures of economic performance. Additional experiments examined the stability of the results across firm characteristics, investment conditions, and alternative assumptions about digital twin efficacy.

3.1 Multi-Objective Optimization Procedure

The investment problem defined in Section 2 was solved using the Non-dominated Sorting Genetic Algorithm II (NSGA-II) developed by Deb et al. (2002). The algorithm was selected to identify portfolios that balance expected economic return against downside risk without imposing a fixed preference weight between the two objectives. Each candidate solution specified the projects selected for funding, the initial digital twin adoption decision, and the threshold governing a subsequent upgrade.

The search used a population of 100 candidate solutions over 100 generations. Binary project-selection and deployment variables were updated through uniform crossover and bit-flip mutation, while the continuous upgrade threshold was modified using blend crossover and Gaussian mutation operators. Feasibility was handled through constrained domination, giving priority to solutions satisfying the investment budget before comparing their objective values. Among feasible solutions, non-dominated sorting and crowding distance guided selection towards portfolios with favourable risk–return characteristics and adequate coverage of the Pareto frontier.

For each firm, candidate solutions were evaluated using a common set of 3,000 Monte Carlo scenarios. The same underlying stochastic draws were used when comparing alternative decisions, reducing variation attributable to scenario sampling. Each scenario included the systematic factor, stage-specific disturbances, revenue shocks, and digital twin efficacy. Project development stages were simulated sequentially, with expenditure terminating when a stage failed. Successful projects generated commercial revenue over the specified operating period, and complementary returns were included when both projects in a synergy pair completed their development stages.

Digital twin deployment modified the project outcomes and cash flows through the four channels defined in Table 2. When the lite system was selected, its initial capital cost was incurred at the beginning of the investment horizon. The efficacy signal was observed at the end of year 1, after which the upgrade decision was determined by the selected threshold. An upgrade incurred the additional capital expenditure and applied the full-level intervention parameters to subsequent cash flows and project stages.

Scenario-level cash flows were discounted according to Eq. (5). The resulting NPV distribution was used to estimate expected NPV and the CVaR of portfolio loss, as specified in Eqs. (6) and (7). The optimization therefore evaluated each candidate through both its expected economic performance and its exposure to adverse outcomes.

Algorithm 1. Evaluation of a candidate decision and the NSGA-II search.

Input: one firm instance (Table A1); 3,000 common-random-number scenarios for the search and 20,000 independent scenarios for validation.

1. Initialise a population of 100 candidate decisions (x, d_0, κ): x selects the projects, d_0 deploys the lite twin and κ is the upgrade threshold.
2. Project selection. Check the funded set x against the budget, Eq. (8); rank infeasible candidates by constraint violation (constrained domination).
3. Twin deployment. If $d_0 = 1$, pay the lite capex at $t = 0$ and apply the lite-level effects of Table 2, scaled by θ; the factor loading ρ follows Eq. (3).
4. Scenario simulation. For each scenario m, draw Z, the idiosyncratic terms and the revenue shocks; run the stage gates of each funded project in sequence with the twin-adjusted success probability and stage cost; a failed project stops spending.
5. Signal observation and upgrade. At the end of year 1 observe S, Eq. (4); if $S > \kappa$, pay the upgrade cost (0.65 USD m) and apply full-level effects from then on.
6. Objectives. Compute NPV_m, Eq. (5), and aggregate over scenarios to $f_1 = -\mathrm{E}[\mathrm{NPV}]$ and $f_2 = \mathrm{CVaR}_{5\%}(-\mathrm{NPV})$, Eq. (6).
7. NSGA-II. Rank by constrained non-dominated sorting and crowding distance; create offspring by uniform crossover and bit-flip mutation on (x, d_0) and blend/Gaussian operators on κ; repeat steps 2 to 6 for 100 generations.
8. Validation. Re-evaluate every frontier point on the 20,000 validation scenarios and compute the hypervolume against the firm's reference point and the gain in E[NPV] at matched CVaR.

3.2 Benchmark Investment Strategies

Six investment strategies were compared to distinguish the value of digital twin adoption from the value of evaluating the technology within the innovation portfolio.

The integrated strategy (A) jointly optimized project selection, initial digital twin deployment, and the upgrade threshold. It represented the full TEIP decision problem and served as the principal comparison strategy.

The decoupled strategy (B) evaluated the digital twin independently of the development portfolio. Adoption was approved when the best stand-alone expected NPV of the twin, including its upgrade decision, was positive. The innovation portfolio was then optimized conditional on that technology decision. This procedure represented an investment appraisal in which the direct economic case for the technology was established before project allocation.

The portfolio-first sequential strategy (B2) reversed the order of appraisal. It first constructed the optimized no-twin portfolio frontier and then evaluated the addition of a lite twin to each selected portfolio. Three candidate upgrade thresholds were examined, and adoption was permitted when the budget constraint was satisfied. The project selection was not subsequently changed. This benchmark measured the value of recognizing the digital twin's effects on an existing optimized portfolio without undertaking full joint re-optimization.

The no-twin strategy (C) optimized the innovation portfolio without digital twin investment. It provided the principal reference for measuring the economic contribution of adopting the technology.

Two conventional selection procedures were also included. The mean–variance strategy (D) used hill-climbing to optimize a weighted combination of expected NPV and standard deviation over eleven values of the risk preference parameter λ, without digital twin adoption. The greedy strategy (G) ranked projects according to stand-alone expected NPV per unit of investment outlay and selected projects in that order subject to the budget constraint.

The first four strategies formed a sequence of increasingly integrated investment decisions. Strategy C excluded the technology, strategy B assessed its direct economic case before portfolio selection, strategy B2 evaluated it against an existing optimized portfolio, and strategy A selected the technology and projects jointly. Comparing these strategies separated the contribution of portfolio-aware technology appraisal from the additional return associated with reselecting projects after the digital twin decision.

The mean–variance and greedy strategies provided supplementary references for conventional investment selection. Because these procedures differed from the main strategies in their optimization methods and, in the mean–variance case, their risk measure, their results were interpreted as comparisons between complete decision procedures rather than as isolated estimates of digital twin value.

3.3 Simulation Settings and Performance Measures

The main experiments used 30 simulated firms with different innovation portfolios and operating characteristics. Project sets were generated using distinct random seeds, while the firm-level parameters followed the distributions in Table A1. The baseline defect rate was drawn from $q \sim U$ (3%, 15%), annual production value from $V \sim U$ (3, 10)

USD million, and the systematic dependence parameter from $\rho_0 \sim U$ (0.2, 0.5). Digital twin efficacy uncertainty also varied across firms, with its coefficient of variation drawn from the specified baseline range.

The economic performance of each strategy was assessed using hypervolume and expected NPV at matched CVaR levels. Hypervolume measures the area dominated by the non-dominated solutions relative to a reference point in the objective space and provides a measure of multi-objective search performance (Zitzler & Thiele, 1999; Zitzler et al., 2003). It therefore captures both the economic quality of the solutions and their coverage of the risk–return frontier. A separate reference point was used for each firm, consistent with the treatment of hypervolume in multi-objective optimization (Zitzler et al., 2003).

For two investment strategies A and B, the relative hypervolume gain was calculated as

$$\Delta \mathrm{HV_k} = \frac{\mathrm{HV}_k(F_\mathrm{A}) - \mathrm{HV}_k(F_\mathrm{B})}{\mathrm{HV}_k(F_\mathrm{B})} \times 100\% \tag{11}$$

where, F_A and F_B denote the independently evaluated frontiers of strategies A and B, respectively, and HV_k denotes hypervolume for firm k under its reference point.

Because hypervolume depends on the reference point, a second comparison was made using expected NPV at matched CVaR levels. Twenty risk levels were used to compare the expected economic returns attainable by the alternative strategies under equivalent downside-risk conditions. This measure expressed the investment difference in monetary terms and provided a complementary assessment of the frontier comparisons.

The statistical analysis was conducted across paired firm instances. Differences between strategies were assessed using the Wilcoxon signed-rank test, with bootstrap confidence intervals calculated from 2,000 resamples. The nine principal comparisons, comprising four investment-strategy contrasts and five channel-ablation contrasts, were adjusted using the Holm (1979) procedure. Matched-pairs rank-biserial correlations were reported to describe the direction and magnitude of paired differences (Kerby, 2014), while percentile bootstrap intervals followed Efron (1979).

The paired design ensured that comparisons were made between investment strategies applied to the same generated firms. Statistical results therefore described the consistency of economic differences across the simulated firm population.

3.4 Sensitivity, Robustness, and Validation Tests

The experimental programme included sensitivity analysis, convergence assessment, independent population studies, and tests of investment performance under alternative economic conditions.

Global sensitivity analysis. Variance-based sensitivity analysis was conducted using the first-order and total-effect Sobol indices (Sobol', 2001; Saltelli et al., 2010). Ten input parameters were examined using 400 base samples, corresponding to approximately 4,800 model evaluations. The analysis considered two outputs for a fixed risk-aware portfolio: the increase in expected NPV associated with digital twin adoption and the corresponding reduction in CVaR. Bootstrap intervals were calculated for the sensitivity estimates.

Robustness and convergence. The integrated and no-twin strategies were compared across 12 firms under alternative investment conditions. The experiments included budget levels of 30% and 70% of full-portfolio expected outlay, CVaR tail probabilities of 1% and 10%, higher systematic correlation, greater efficacy uncertainty, a pessimistic digital twin specification, and portfolios containing 15 candidate projects. These tests examined how capital availability, risk preference, project dependence, and technology performance affected the economic results.

Optimization convergence was assessed on six firms by increasing the search effort to a population of 200 over 300 generations. The resulting frontiers were compared with those obtained from the baseline configuration to assess the stability of the multi-objective search.

Population-level investment appraisal. A separate sample of 400 simulated firms was generated to examine differences between stand-alone and portfolio-aware digital twin adoption decisions. The population included variation in project characteristics and a cost-scale parameter ranging from 0.8 to 1.4. For each firm, hill-climbing was used to maximise the risk-adjusted utility $U = \mathrm{E[NPV]} - 0.5\mathrm{CVaR}_{0.05}(-\mathrm{NPV})$ under both no-twin and digital twin investment conditions. The digital twin evaluation considered upgrade thresholds of 0.8 and 1.2 and used 1,500 scenarios per firm.

The portfolio-aware economic gain was defined as the difference in optimized utility between the two investment conditions, denoted by ΔU. To measure the contribution beyond stand-alone appraisal, the Twin Externality Index was defined as

$$\mathrm{TEI} = \Delta \mathrm{U} - \max(\mathrm{E_{SA}}, 0) \tag{12}$$

where, $\mathrm{E_{SA}}$ denotes the best stand-alone expected NPV of the digital twin evaluated without development projects. The index measured the difference between the portfolio-aware utility gain and the positive value credited by

independent appraisal. An investment decision was classified as misappraised when the stand-alone and portfolio-aware adoption rules produced different outcomes.

Value of staged learning. The economic value of the upgrade option was examined separately for ten firms with fixed innovation portfolios. The analysis compared a signal-based upgrade policy with the better of two static alternatives: retaining the lite twin or upgrading without conditioning the decision on the efficacy signal.

For a fixed portfolio, the value of flexibility was defined as

$$\Lambda = \max_{\kappa} \mathrm{E}\left[NPV(\kappa)\right] - \max\left\{\mathrm{E}[\mathrm{NPV}_{\mathrm{lite}}], \mathrm{E}\left[\mathrm{NPV}_{\mathrm{upgrade}}\right]\right\} \tag{13}$$

The upgrade threshold was selected using one set of scenarios and evaluated using an independent set. Efficacy uncertainty was varied over coefficients of variation from 0.1 to 1.8, while signal noise ranged from 0.05 to 3.0. The comparison was conducted on a risk-neutral expected-NPV basis to isolate the economic value of using the efficacy signal to determine the upgrade decision.

Independent confirmation and hold-out evaluation. The principal integrated-versus-no-twin comparison was repeated on 100 additional firms drawn from the baseline parameter ranges using new random seeds. A further 30 hold-out firms were generated from wider ranges: defect rates of 1%–20%, annual production values of USD 1.5–14 million, systematic correlation parameters of 0.1–0.7, efficacy coefficients of variation of 0.2–1.2, and cost-scale parameters of 0.7–1.8. These firms contained 8–14 candidate projects and budget levels of 30%, 50%, or 70%.

The extended ranges examined investment performance across a broader combination of production scale, project characteristics, technology costs, and capital availability.

Efficacy heterogeneity and spill-over effects. The baseline model applied a common efficacy factor to the digital twin's intervention channels. An additional experiment allowed project-specific efficacy to vary around the firm-level factor through a lognormal multiplier. The dispersion parameter took values of 0, 0.3, 0.6, and 1.0. The spill-over weight applied to non-process projects was separately varied between 0, 0.3, and 0.6. Each setting was evaluated on 12 firms.

These experiments examined the economic consequences of differences in technological effects across projects and of restricting digital twin influence to process-related activities.

Downside-risk assessment. Portfolio risk was further examined through the probability of negative NPV, expected shortfall, CVaR, and the fifth percentile of the NPV distribution. The integrated and no-twin strategies were compared at a common expected-return target equal to 80% of the maximum return on the no-twin frontier. This comparison assessed downside protection at a comparable economic return rather than solely through aggregate frontier measures.

Technology investment policy analysis. A separate policy experiment considered 60 simulated firms choosing digital twin adoption and innovation projects under alternative public support measures. Each firm maximised a risk-adjusted objective of the form $\mathrm{E}[\mathrm{NPV}] - \lambda_i \mathrm{CVaR}$, using hill-climbing to select the investment strategy. Capital subsidies ranging from 0% to 60% and research and development tax credits ranging from 0% to 20% of project costs were examined.

The policy analysis also included a social-value proxy that added a knowledge spill-over equal to 30% of expected project revenue to private NPV before fiscal transfers. The experiment compared technology adoption, public expenditure, and the allocation of support between firms that adopted because of the policy and those that would have adopted without it.

Taken together, these experiments provided a consistent basis for evaluating digital twin investment across individual portfolios, heterogeneous firms, alternative technology conditions, and policy settings. The resulting economic comparisons are presented in Section 4.

4. Results and Economic Evaluation

The economic performance of digital twin investment was examined through comparisons of alternative investment strategies, decomposition of the sources of value, sensitivity analysis, and adoption threshold calculations. The analysis then considered differences in investment decisions across heterogeneous firms, the value of staged upgrading, downside-risk performance, and the consequences of technology investment policies. Unless otherwise indicated, the principal portfolio comparisons were evaluated on 20,000 independent Monte Carlo scenarios for each firm.

4.1 Economic Performance of Alternative Investment Strategies

The integrated investment strategy produced higher risk–return frontier hypervolume than the principal benchmarks across the 30 simulated firms. Relative to portfolios without digital twins, the mean hypervolume gain was 26.1%, with a median of 22.6% and a 95% bootstrap confidence interval of 18.3%–35.9%. The integrated

strategy also exceeded the decoupled investment procedure by 6.2% and the portfolio-first sequential procedure by 1.7% (Table 3).

Table 3. Out-of-sample hypervolume gains of the integrated investment strategy across 30 simulated firms

Comparison	Mean Hypervolume Gain	95% Bootstrap CI	Firms With Higher Hypervolume
Integrated vs. no twin	26.1% (median 22.6%)	18.3%–35.9%	30/30
Integrated vs. decoupled	6.2% (median 4.1%)	3.8%–8.7%	30/30
Integrated vs. portfolio-first sequential	1.7% (median 1.4%)	1.3%–2.2%	29/30
Integrated vs. mean–variance	486%	312%–697%	30/30

Note: CI = confidence interval. The integrated, no-twin, decoupled, and portfolio-first strategies used the same NSGA-II procedure, expected NPV–CVaR objectives, and independent scenario evaluation. The mean–variance benchmark used hill-climbing, eleven risk-preference settings, and standard deviation as its risk measure, without digital twin investment. Its hypervolume difference therefore reflects several methodological and investment-design differences.

The comparisons reveal two distinct sources of economic improvement. The first arises from recognising the effect of digital twin adoption on the innovation portfolio. The second arises from reconsidering project selection after the technology decision. The portfolio-first strategy already produced a 24.0% hypervolume gain over the no-twin benchmark, compared with 26.1% under joint optimization. Because these gains are relative ratios, the additional contribution of joint optimization is multiplicative: $1.240 \times 1.017 \approx 1.261$. Thus, most of the improvement was obtained by evaluating the digital twin within an optimized portfolio, while joint project reselection provided a further gain.

The stand-alone appraisal rule approved digital twin adoption in 16 of the 30 firms, equivalent to 53%. By comparison, digital twins appeared in an average of 71% of the solutions along the integrated frontiers. These measures describe different decision levels: the first concerns firm-level approval under a stand-alone rule, while the second concerns the proportion of frontier solutions containing the technology. Their difference illustrates the greater role assigned to digital twin investment when its effects on development projects are included.

The economic improvement was also evident when the strategies were compared at equivalent downside-risk levels. The integrated frontier generated an average increase of USD 0.75 million in expected NPV relative to the no-twin frontier. At the highest-return end, expected NPV increased by USD 0.74 million, corresponding to 11.2%, while CVaR declined from USD 3.19 million to USD 2.84 million. The integrated strategy therefore combined higher expected economic returns with lower tail exposure.

Figure 2 presents the independently evaluated frontiers for two representative firms. The integrated solutions occupied more favourable regions of the expected NPV–CVaR space than the alternative investment procedures. The comparison with mean–variance optimization was particularly large, although this difference combined the effects of technology adoption, optimization method, frontier resolution, and risk measure. The more directly comparable no-twin benchmark isolated the contribution of including digital twin investment within the same multi-objective framework.

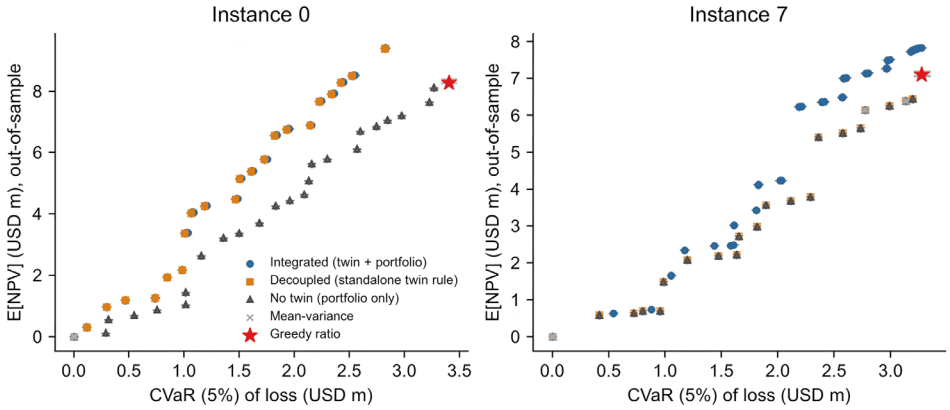


Figure 2. Out-of-sample NPV–CVaR frontiers for two representative firms under alternative investment strategies

Note: NPV = net present value; CVaR = conditional value-at-risk; USD m = USD million.

The improvement varied across firms. Figure 3 shows that the integrated-versus-no-twin hypervolume gain increased with the baseline defect rate, with a correlation coefficient of 0.73. The correlation between defect rate and the proportion of integrated frontier solutions containing a digital twin was 0.50. Firms facing greater recoverable production losses therefore had stronger economic incentives to adopt the technology.

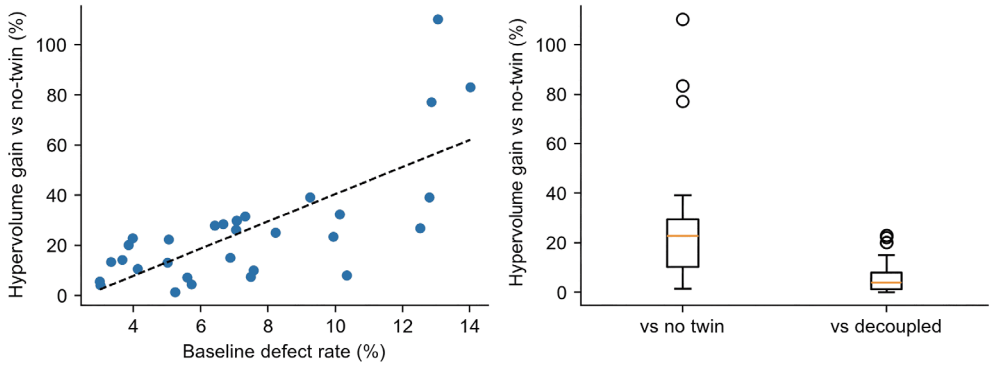


Figure 3. (a) Hypervolume gain versus baseline defect rate; (b) distribution of hypervolume gains across 30 firms

The independent evaluation also indicated stable optimization performance. The ratio of out-of-sample to in-sample expected return was 1.008 for the integrated strategy and 1.011 for the no-twin strategy, indicating limited selection effects associated with the optimization scenarios (Smith & Winkler, 2006). Increasing the optimization effort to a population of 200 over 300 generations changed hypervolume by an average of 0.04%, with a maximum change of 0.11%.

The paired Wilcoxon tests supported the direction of the principal comparisons, with p-values below 10^{-8}. The integrated strategy exceeded the no-twin and decoupled strategies in all 30 firms and the portfolio-first strategy in 29. These results establish the economic distinction between evaluating a digital twin independently and valuing it as part of the development portfolio.

4.2 Sources of Digital Twin Investment Value

The integrated valuation model included four channels through which digital twin adoption affected economic performance: direct operating savings, higher project success probabilities, lower development costs, and reduced dependence between project outcomes. The contribution of staged upgrading was examined separately. Channel-ablation experiments measured the change in portfolio performance when each mechanism was excluded.

Table 4 reports the results of these experiments. The full model produced a mean hypervolume gain of 26.8% over the no-twin benchmark. This was a separate experimental evaluation from the principal comparison in Table 3, which reported 26.1%; the ablation results were assessed against their own full-model reference.

Table 4. Contribution of digital twin investment channels to portfolio performance across 30 simulated firms

Model Specification	Hypervolume Gain Over No Twin	Expected NPV Gain at Matched CVaR (USD Million)
Full model	26.8 ± 8.6%	0.59
Without decorrelation	30.0 ± 8.9%	0.66
Without development cost reduction	18.0 ± 7.9%	0.41
Without stage-success improvement	13.0 ± 7.6%	0.31
Direct operating savings only	10.1 ± 7.0%	0.24
Without upgrade option	25.5 ± 8.1%	0.56

Note: NPV = net present value; CVaR = conditional value-at-risk. Hypervolume values are mean gains with 95% confidence-interval half-widths. The ablation experiments use their own full-model reference.

The largest contribution arose from changes in project success probabilities. Removing the success-improvement channel reduced the hypervolume gain from 26.8% to 13.0%, a difference of 13.8 percentage points. Excluding development cost reduction lowered the gain to 18.0%, corresponding to an 8.8-point difference based on the rounded table values. These results indicate that the economic value of digital twin adoption depended strongly on its influence over the development process, rather than exclusively on production cost savings.

The direct-savings-only specification retained a hypervolume gain of 10.1%. Relative to the full-model gain of 26.8%, this represented approximately 38% of the observed improvement. The remaining 62% was associated with the combined effects of the development-project channels. This proportion is a residual comparison between model specifications rather than an additive allocation of value to independent mechanisms. The channels interact through project selection, development outcomes, and portfolio constraints.

The staged upgrade also contributed to investment performance. Removing the upgrade option reduced the gain from 26.8% to 25.5%, a difference of 1.3 percentage points. The option allowed the firm to postpone the larger

capital commitment until information about technology efficacy became available. Its contribution was smaller than that of project success and development cost reduction, but it added value to the staged investment strategy.

The decorrelation channel produced a different result. Excluding this channel increased the hypervolume gain from 26.8% to 30.0%. For a fixed portfolio, reduced dependence between stage outcomes lowered CVaR by approximately USD 0.01 million but also reduced expected synergy returns, resulting in a USD 0.11 million decline in expected NPV. The change in joint project success was therefore economically important when complementary projects generated additional revenue.

This result follows the mechanism established in Proposition 2. Lower dependence reduces the concentration of certain joint outcomes, but it also changes the probability that complementary projects complete successfully together. In the simulated portfolios, the reduction in complementary returns exceeded the associated improvement in downside risk. The corresponding channel-ablation comparisons are presented in Figure 4.

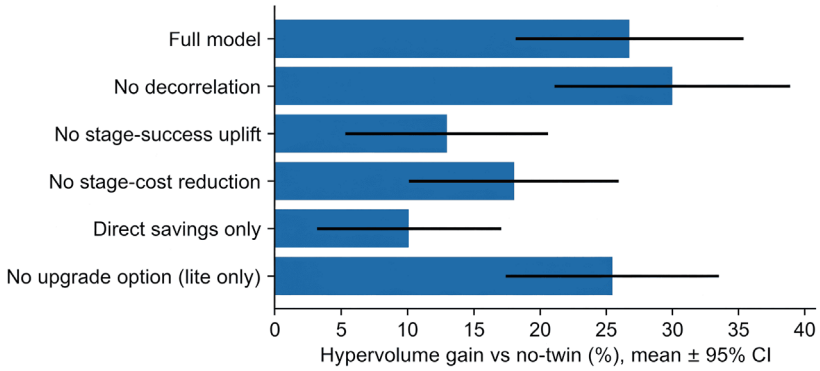


Figure 4. Hypervolume gains under alternative digital twin channel specifications (95% confidence intervals)
Note: CI = confidence interval.

The ablation comparisons show that the main economic contribution of the technology arose from changes in the development portfolio. Improving stage completion and reducing development expenditure generated larger gains than direct savings alone. The staged upgrade added further value, while the correlation channel demonstrated that a reduction in joint failure exposure does not necessarily translate into a higher expected portfolio return.

4.3 Sensitivity and Robustness of Investment Outcomes

The economic performance of the integrated strategy was examined under alternative budget levels, risk measures, project characteristics, and digital twin efficacy conditions. Table 5 reports the integrated-versus-no-twin hypervolume gains for 12 firms under each setting.

Table 5. Robustness of the integrated investment strategy under alternative economic conditions

Setting	Hypervolume Gain (95% CI)	Firms With Gain	Expected NPV Gain (USD Million)
Base case	16.6% (11.2–22.6)	12/12	0.49
Budget at 30%	13.5% (6.5–22.3)	12/12	0.23
Budget at 70%	18.3% (13.7–23.4)	12/12	0.72
CVaR tail level 1%	12.8% (8.5–17.6)	12/12	0.39
CVaR tail level 10%	22.6% (16.4–29.0)	12/12	0.66
High systematic correlation (ρ_0 = 0.6)	10.5% (5.9–15.7)	12/12	0.37
High efficacy uncertainty (CV = 1.0)	12.1% (8.5–16.2)	12/12	0.36
15 candidate projects	19.9% (14.8–25.4)	12/12	0.86
Pessimistic twin specification	0.0%	—	0.00

Note: The pessimistic specification used an efficacy level of 0.6, a 0.6 multiplier on technology effects, and a 1.5 multiplier on costs. The digital twin was declined in 11 of 12 firms. The em dash (—) indicates that no firms achieved a hypervolume gain under this setting, as the resulting investment frontiers coincided with the no-twin benchmark. CI = confidence interval; NPV = net present value; CVaR = conditional value-at-risk; CV = coefficient of variation.

The integrated strategy generated positive hypervolume gains in all 12 firms under the principal alternative settings. Increasing the budget from 30% to 70% raised the mean gain from 13.5% to 18.3%, while the

corresponding expected NPV improvement increased from USD 0.23 million to USD 0.72 million. Greater capital availability allowed firms to finance a broader set of development projects alongside the digital twin.

The choice of tail-risk level also affected the investment frontier. The mean hypervolume gain was 12.8% at the 1% tail level and 22.6% at the 10% level. The larger tail probability placed less emphasis on the most extreme losses, allowing greater scope for portfolios with higher expected returns.

Higher systematic correlation reduced the mean gain to 10.5%, while greater efficacy uncertainty reduced it to 12.1%. Expanding the candidate project set to 15 increased the gain to 19.9%, with an expected NPV improvement of USD 0.86 million. A larger development pipeline provided more opportunities for the technology to affect project selection and expected returns.

Under the pessimistic technology specification, the optimiser declined digital twin adoption in 11 of 12 firms. The resulting investment frontiers coincided with the no-twin benchmark in those cases. The framework therefore selected the technology according to its economic contribution rather than treating adoption as a predetermined outcome.

Global sensitivity analysis further identified the parameters most closely associated with investment value. The first-order Sobol indices for the expected NPV gain were highest for baseline defect rate (0.31), mean technology efficacy (0.21), and annual production value (0.16). Their combined first-order contribution was 0.68, indicating that these three variables accounted for approximately 68% of the output variance in the specified sensitivity experiment.

For CVaR reduction, the leading first-order indices were defect rate (0.31), production value (0.19), technology cost scale (0.12), and mean efficacy (0.12). The corresponding total-effect indices were 0.38, 0.27, 0.20, and 0.19. Systematic correlation, efficacy uncertainty, discount rate, and revenue scale each had total-effect indices below 0.10. The first-order and total-effect sensitivity indices are shown in Figure 5.

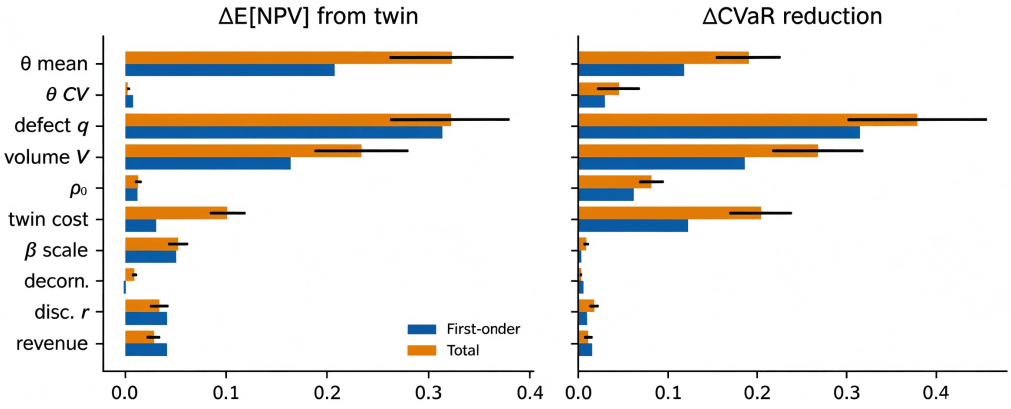


Figure 5. First-order and total-effect Sobol indices for expected NPV gain and CVaR reduction

Note: The horizontal axis represents the Sobol sensitivity index. The variables on the vertical axis denote the baseline defect rate (q), annual production value (V), mean technology efficacy (θ), coefficient of variation of efficacy (CV), baseline systematic correlation (ρ_0), technology cost scale, success uplift coefficient (β), decorrelation coefficient, discount rate (r), and revenue scale. First-order indices measure the direct variance contribution of each parameter, whereas total-effect indices capture both direct effects and interactions with other parameters. NPV = net present value; CVaR = conditional value-at-risk.

These results identify recoverable production losses and technology efficacy as the principal drivers of the investment decision. The defect rate and production value determine the scale of direct operating savings, while efficacy affects both operational and development-project channels. Together, they provide the strongest basis for the initial economic assessment of digital twin adoption.

4.4 Investment Break-Even Thresholds and Adoption Decisions

The investment model was used to derive practical adoption thresholds based on recoverable scrap value. This measure, defined as the product of annual production value V and defect rate q, links the operating conditions of a manufacturing firm to the economic return available from digital twin investment.

An 8 × 8 grid was constructed using defect rates from 2% to 16% and annual production values from USD 2 million to USD 12 million. Under stand-alone appraisal, digital twin NPV was positive in 38 of the 64 combinations. Under portfolio-aware evaluation, the integrated risk-adjusted utility gain was positive in 55 combinations. The corresponding adoption regions covered approximately 59% and 86% of the grid.

For the reference firm, the adoption rule was expressed as

$$Vq > \tau, \tau_{\mathrm{SA}} \approx 0.39, \tau_{\mathrm{INT}} \approx 0.17 \text{ USD million/year} \tag{14}$$

The stand-alone threshold follows the closed-form relationship in Proposition 1, while the portfolio-aware threshold incorporates the economic contribution of development projects. The latter was approximately 56% lower for the reference portfolio.

The stand-alone threshold is determined by technology ownership costs, expected efficacy, and the discount factor. Table 6 shows its response to changes in investment costs and operating effectiveness.

Table 6. Stand-alone break-even recoverable scrap value for the lite twin (USD million/year)

Direct-Savings Multiplier	Base Ownership Cost	Cost +25%	Cost +50%
0.6	0.64	0.80	0.96
0.8	0.48	0.60	0.72
1.0	0.38	0.48	0.58
1.2	0.32	0.40	0.48

At the reference efficacy level, the break-even recoverable scrap value was approximately USD 0.384 million per year. When direct-savings effectiveness declined to 60% of the reference level and ownership costs increased by 50%, the threshold rose to approximately USD 0.96 million per year. The investment decision was therefore sensitive to both the expected operating effect and the total cost of ownership.

The economic threshold also depended on the implementation level. Table 7 compares the lite twin, immediate full deployment, and the incremental upgrade from lite to full.

Table 7. Direct-savings break-even conditions for alternative digital twin investment options

Investment Option	Capital Cost (USD Million)	Annual Operating Cost (USD Million)	PV of Ownership Cost (USD Million)	Break-Even Recoverable Scrap Value (USD Million/Year)
Lite twin at year 0	0.25	0.03	0.410	0.384
Full twin at year 0	0.80	0.10	1.333	0.555
Upgrade at end of year 1	0.65	0.07 incremental	0.991 at year 1	0.814, expressed as ($Vq\theta$)

Note: PV = present value. The annuity factors are ($A_8 = 5.335$) and ($A_7 = 4.868$) at a 10% discount rate. The upgrade calculation considers additional operating savings in years 2–8 and excludes development-project effects.

Immediate full deployment required a higher direct-savings threshold than the lite system because of its larger ownership cost. The incremental upgrade required ($Vq\theta$) above approximately USD 0.814 million per year when evaluated through additional operating savings alone. Across the baseline parameter ranges, recoverable scrap value extended from approximately USD 0.09 million to USD 1.5 million annually. At unit efficacy, about 22% of the sampled draws exceeded the upgrade threshold. For other firms, development-project effects and the information obtained during initial deployment contributed to the economic case for expansion.

The ownership horizon and discount rate also affected the adoption threshold. Table 8 reports the stand-alone break-even values for alternative useful lives and discount rates.

Table 8. Stand-alone break-even recoverable scrap value by discount rate and useful life (USD million/year)

Discount Rate	5 Years	8 Years	10 Years
8%	0.463	0.368	0.336
10%	0.480	0.384	0.353
12%	0.497	0.402	0.371

At a 10% discount rate, shortening the useful life from eight to five years raised the threshold from 0.384 to 0.480 USD million per year, an increase of 25%. Raising the discount rate from 10% to 12% at an eight-year life increased the threshold to 0.402, approximately 4.7% above the reference value. The useful life therefore had a stronger influence over the tested range than the discount rate.

The threshold equation can also be used to calculate the mean efficacy required for direct operating savings to recover the ownership cost. Table 9 reports this relationship for different recoverable scrap values.

A firm with annual recoverable scrap value of USD 0.17 million would require a mean efficacy of approximately 2.26 for the lite twin to recover its ownership cost through direct savings alone. This corresponds to recovering approximately 45% of the scrap value, compared with the 20% reference recovery rate. The difference illustrates the economic contribution required from the development portfolio when direct savings do not independently justify adoption.

Table 9. Required mean efficacy and scrap recovery rate for stand-alone lite-twin investment

Recoverable Scrap Value (USD Million/Year)	Required Mean Efficacy	Required Scrap Recovery	Investment Position
0.10	3.84	77%	Below both thresholds
0.17	2.26	45%	At the integrated threshold
0.20	1.92	38%	Between thresholds
0.30	1.28	26%	Between thresholds
0.384	1.00	20%	At the stand-alone threshold
0.60	0.64	13%	Above both thresholds
1.00	0.38	8%	Above both thresholds

The portfolio-aware threshold provides a complementary investment measure. Applying Corollary 1 to the reference thresholds gives an externality-equivalent present value of approximately USD 0.23 million, representing about 56% of the lite twin's ownership cost. This amount expresses the portfolio contribution implied by the threshold difference under the expected-value assumptions of the corollary. The stand-alone and portfolio-aware break-even regions are illustrated in Figure 6.

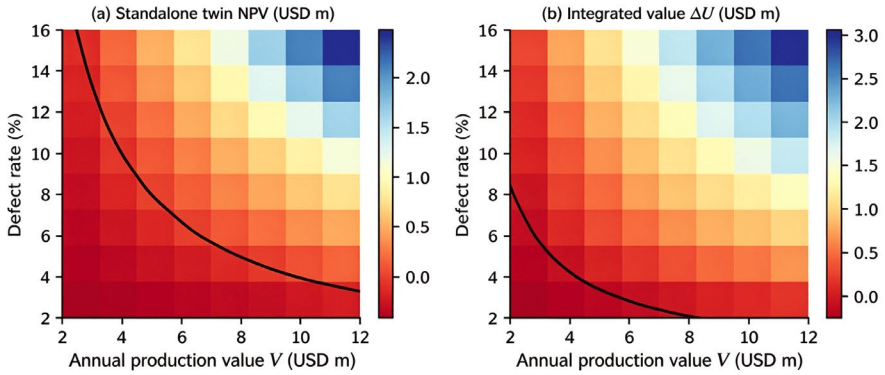


Figure 6. Break-even maps: (a) stand-alone NPV; (b) portfolio-aware utility gain
Note: NPV = net present value.

The two thresholds serve different purposes. The stand-alone threshold is a direct-savings benchmark that can be recalculated from a firm's technology costs and operating conditions. The integrated threshold also depends on its development portfolio, project interactions, and capital allocation opportunities. The population analysis in Section 4.5 examines how these portfolio characteristics affect adoption decisions across heterogeneous firms.

4.5 Portfolio Externalities and Investment Misappraisal

The population analysis extended the comparison between stand-alone and portfolio-aware investment appraisal to 400 simulated firms. The firms differed in production conditions, development portfolios, technology costs, and project dependence. The objective was to measure how frequently the two appraisal procedures selected different adoption decisions and to quantify the associated economic consequences.

Stand-alone appraisal approved digital twin adoption in 58.5% of firms, while portfolio-aware appraisal selected adoption in 70.5%. The decisions differed for 24.0% of firms. In 18.0% of cases, stand-alone appraisal rejected a digital twin that was selected under portfolio-aware evaluation. In a further 6.0%, stand-alone appraisal approved adoption while the portfolio-aware decision favored alternative uses of the investment budget.

The adoption decisions and associated utility differences are summarized in Table 10.

Table 10. Differences between stand-alone and portfolio-aware investment appraisal across 400 simulated firms

Measure	Value
Adoption under stand-alone appraisal	58.5%
Adoption under portfolio-aware appraisal	70.5%
Firms with different adoption decisions	24.0%
Stand-alone rejection, portfolio-aware adoption	18.0%
Stand-alone adoption, portfolio-aware rejection	6.0%
Mean utility gain among portfolio-aware adopters	USD 1.10 million
Mean Twin Externality Index among portfolio-aware adopters	USD 0.60 million
Mean utility forgone by following stand-alone appraisal	USD 0.084 million per firm

The average utility loss was USD 0.47 million among firms where stand-alone appraisal rejected an economically preferred investment and USD 0.36 million among firms where it approved an investment displaced by better portfolio opportunities. Across the population, following the stand-alone rule resulted in an average forgone utility of USD 0.084 million per firm, equivalent to approximately 11% of the total gain available from the technology.

The Twin Externality Index averaged USD 0.60 million among firms selecting adoption under portfolio-aware evaluation, with a median of USD 0.55 million. This measure quantified the economic value attributed to the development portfolio beyond the positive stand-alone expected NPV credited to the digital twin.

The stand-alone adoption decision was closely related to the ratio of recoverable scrap value to technology cost scale. A threshold of approximately 0.39 on this ratio reproduced 99.8% of the stand-alone decisions. Portfolio-aware adoption depended on additional firm characteristics, particularly the proportion of process-related projects and the dependence between project outcomes.

Table 11 reports adoption rates and mean utility differences across quintiles of the scale-adjusted recoverable scrap value ratio.

Table 11. Digital twin adoption by quintile of the scale-adjusted recoverable scrap value ratio

Hurdle-Ratio Quintile	Firms	Stand-Alone Adoption	Portfolio-Aware Adoption	Mean (ΔU) (USD Million)
0.05–0.23	80	0%	30%	−0.35
0.23–0.38	80	0%	55%	0.03
0.38–0.58	80	92%	81%	0.44
0.58–0.87	80	100%	88%	0.92
0.87–1.89	80	100%	99%	1.98

Note: Mean (ΔU) describes the reported utility difference associated with digital twin investment across firms in each quintile, rather than the utility gain conditional on adoption.

In the lowest two quintiles, stand-alone appraisal rejected the digital twin for every firm, whereas portfolio-aware appraisal selected adoption for 30% and 55%, respectively. The opposite pattern occurred in parts of the higher-ratio groups, where some firms had sufficient direct savings to justify the technology independently but obtained greater portfolio value by allocating capital to other projects.

A single threshold classified 79% of portfolio-aware adoption decisions correctly. Including process-project share and systematic correlation increased classification accuracy to 84%. These results show that recoverable scrap value provides an effective initial screening variable, while portfolio characteristics determine the investment decision for firms close to the adoption boundary.

A regression analysis of the portfolio-aware utility gain on eight standardized firm characteristics explained 70% of its variation. Recoverable scrap value had the largest positive coefficient, at USD 0.83 million per standard deviation (95% bootstrap CI 0.76–0.89). Process-project share contributed USD 0.18 million (0.13–0.23), while technology cost scale and systematic correlation had coefficients of −0.27 (−0.33 to −0.21) and −0.16 (−0.22 to −0.10), respectively.

The proportion of process-related projects also differentiated the two appraisal methods. Across quartiles of process-project share, stand-alone adoption remained between 58% and 60%, while portfolio-aware adoption increased from 65% to 77%. The mean utility gain increased from USD 0.42 million to USD 0.78 million. These results connect the adoption decision directly to the development activities supported by the technology.

Figure 7 presents the adoption patterns across recoverable scrap value quintiles and the principal determinants of portfolio-aware utility gain.

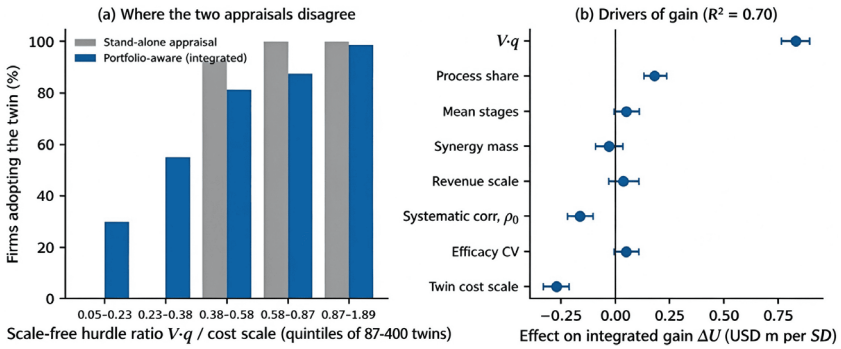


Figure 7. (a) Digital twin adoption rates across 400 firms and (b) determinants of portfolio-aware utility gain

Note: V·q = product of annual production value and defect rate; ΔU = portfolio-aware utility gain; R^2 = coefficient of determination; ρ_0 = systematic correlation parameter; CV = coefficient of variation; SD = standard deviation. Error bars in panel (b) represent 95% confidence intervals.

The population results establish the economic relevance of portfolio-level valuation beyond the main 30-firm experiment. Stand-alone appraisal omitted part of the technology's contribution to development projects and, in other cases, failed to account for the opportunity cost of allocating scarce capital to the digital twin. Portfolio-aware evaluation incorporated both effects within the investment decision.

4.6 Economic Value of Staged Investment and Learning

The staged investment strategy allowed firms to adopt a lite digital twin and decide whether to upgrade after observing an efficacy signal. The economic value of this flexibility was examined separately from the main portfolio optimization, using ten simulated firms with fixed project portfolios.

The value of flexibility was measured by comparing the expected NPV of the best signal-dependent upgrade policy with the better of two static alternatives: retaining the lite system or upgrading independently of the observed signal. Under a decision set containing both static policies, the optimized value of signal-dependent flexibility is non-negative because the firm can always select the better static action.

When efficacy is known, the signal provides no additional information for selecting between the static alternatives. The same limiting result applies when signal noise makes the observation independent of the underlying efficacy. Between these conditions, the signal can guide the upgrade decision towards the technology level with the higher expected economic return.

Table 12 reports the out-of-sample estimates across different levels of efficacy variation and signal noise.

Table 12. Out-of-sample flexibility value of the staged digital twin upgrade (USD thousand)

Efficacy CV	Noise 0.2	Noise 0.35	Noise 0.7	Noise 1.2	Noise 3.0
0.3	24	12	2	−1	0
0.5	72	56	28	8	−1
0.8	138	123	86	49	−1
1.2	212	201	165	116	7
1.8	282	273	241	189	38

Note: CV = coefficient of variation. Values are means across ten firms, rounded to the nearest USD thousand. Small negative estimates arise from selecting the upgrade threshold on one scenario set and evaluating it on an independent sample. They represent finite-sample estimation differences rather than a negative theoretical value of flexibility.

At the baseline setting of efficacy CV 0.5 and signal noise 0.35, the staged upgrade option had an estimated value of USD 0.056 million. Increasing efficacy variation to CV 1.2 raised this value to USD 0.201 million, while CV 1.8 produced USD 0.273 million. The latter was slightly greater than the USD 0.25 million initial capital cost of the lite system.

The value declined as signal noise increased. At CV 1.8, the estimated value fell from USD 0.282 million at noise 0.2 to USD 0.038 million at noise 3.0. The results therefore linked the economic return from staged investment to both the variation in realised technology efficacy and the information available before the upgrade decision. The effects of efficacy variation and signal noise on the value of staged upgrading are illustrated in Figure 8.

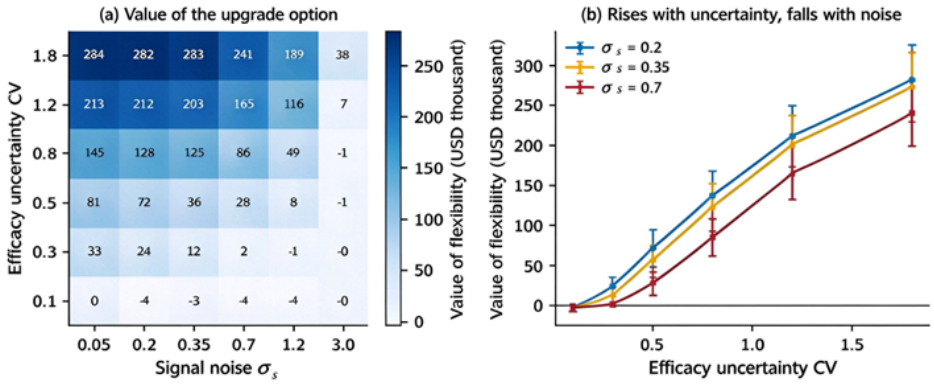


Figure 8. (a) Flexibility value across efficacy and signal noise and (b) flexibility value across efficacy uncertainty

Note: CV = coefficient of variation of technology efficacy; σ_s = standard deviation of the efficacy signal noise. The heatmap in panel (a) displays the value of the upgrade option across different levels of efficacy uncertainty and signal noise, and panel (b) presents the flexibility value across efficacy uncertainty under three distinct signal noise levels (σ_s = 0.2, 0.35, 0.7).

The investment implication is that initial deployment serves two functions. It provides operating and

development-project effects at the lite level while also generating information relevant to the subsequent capital commitment. The value of this information was greatest when technology efficacy varied substantially and the observed signal retained sufficient predictive content.

4.7 Generalisation and Downside Risk Assessment

The principal economic comparisons were repeated on additional firm populations and alternative technology-effect specifications. These experiments examined the consistency of the integrated strategy's performance beyond the initial 30-firm sample.

A confirmation sample of 100 new firms drawn from the baseline parameter ranges produced a mean hypervolume gain of 37.3% over the no-twin benchmark, with a 95% confidence interval of 31.3%–43.6% and a median of 27.0%. The gain was positive in 99 firms and zero in one. Relative to the portfolio-first sequential strategy, joint optimization produced an additional mean gain of 1.7%.

The hold-out sample contained 30 firms drawn from wider ranges of production value, defect rate, technology cost, efficacy variation, and portfolio characteristics. The integrated strategy produced a mean hypervolume gain of 70.1% over the no-twin benchmark, with a median of 40.6% and a 95% confidence interval of 44.3%–98.8%. The gain was positive in 28 firms and zero in two. The larger average improvement reflected the inclusion of firms with higher recoverable scrap values.

The results of the confirmation, hold-out, and technology-effect experiments are summarized in Table 13.

Table 13. Generalization of integrated digital twin investment performance

Evaluation Setting	Gain Over No Twin (95% CI)	Firms With Gain	Gain Over Portfolio-First
Main sample, 30 firms	26.1% (18.3–35.9)	30/30	1.7%
Confirmation sample, 100 firms	37.3% (31.3–43.6)	99/100 (1 tie)	1.7%
Hold-out sample, 30 firms	70.1% (44.3–98.8)	28/30 (2 ties)	1.4%
Project efficacy dispersion 0.0	16.6% (11.1–22.8)	12/12	1.9%
Project efficacy dispersion 0.3	16.5% (11.1–22.7)	12/12	2.0%
Project efficacy dispersion 0.6	16.1% (10.8–22.3)	12/12	2.0%
Project efficacy dispersion 1.0	14.9% (9.7–20.9)	12/12	2.0%
Non-process spill-over 0.0	13.8% (8.4–19.9)	12/12	2.0%
Non-process spill-over 0.3	16.6% (11.1–22.8)	12/12	1.9%
Non-process spill-over 0.6	20.2% (14.7–26.6)	12/12	2.0%

Note: CI = confidence interval. A tie indicates no difference in hypervolume between the two strategies.

The project-specific efficacy experiments examined the effect of relaxing the common-efficacy assumption. Increasing the dispersion parameter from 0 to 1.0 reduced the mean hypervolume gain from 16.6% to 14.9%. The additional gain from joint optimization over portfolio-first evaluation remained approximately two percentage points.

Restricting the technology's effects to process-related projects reduced the gain to 13.8%, while increasing the spill-over weight to 0.6 raised it to 20.2%. The integrated strategy therefore retained an economic advantage when digital twin effects were concentrated in the projects most directly connected to manufacturing processes.

The analytical dependence results were also examined through joint project completion probabilities. Table 14 reports the joint completion probabilities for two three-stage projects and the distribution of completed projects in a ten-project portfolio.

Table 14. Effect of the common dependence parameter on joint project completion

Case	Dependence Parameter	Both Projects Complete	At Most One of Ten Completes	At Least Six of Ten Complete
No twin	0.3500	0.266	0.249	0.369
Lite twin	0.2975	0.244	0.238	0.343
Full twin	0.2275	0.214	0.219	0.303
Dependence floor	0.0350	0.133	0.125	0.136

Note: Values are those reported from numerical integration over the systematic factor. The comparison holds marginal stage-success probabilities fixed at 0.70 and changes the common dependence parameter.

Reducing the dependence parameter from 0.35 to 0.2275 lowered the reported probability of joint completion from 0.266 to 0.214, a reduction of approximately 19%. The probability that at most one of ten projects completed fell by 12%, while the probability that at least six completed fell by 18%. The change therefore affected both adverse and favourable joint outcomes, explaining the importance of project complementarities in the valuation model.

Beyond hypervolume, the analysis examined downside exposure at a common expected-return target equal to 80% of the maximum return on the no-twin frontier. Table 15 reports the resulting portfolio risk measures. Figure 9 illustrates the relationship between the hold-out hypervolume gain and the scale-adjusted recoverable scrap value ratio, together with the changes in downside-risk measures.

Table 15. Downside-risk performance at matched expected return across 30 firms

Risk Measure	No Twin	Integrated	Relative Change	Firms Improved
Probability of negative NPV	20.9%	15.3%	−27%	30/30
Expected shortfall (USD million)	0.359	0.228	−36%	30/30
CVaR 5% (USD million)	2.61	2.12	−19%	28/30
Fifth-percentile NPV (USD million)	−2.22	−1.70	23% reduction in loss magnitude	30/30

Note: NPV = net present value; CVaR = conditional value-at-risk.

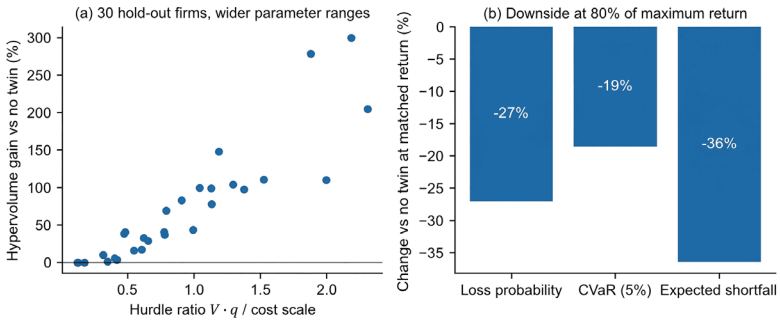


Figure 9. (a) Hold-out hypervolume gains; (b) downside-risk changes at matched expected return

The integrated strategy reduced the mean probability of negative NPV from 20.9% to 15.3%. Expected shortfall declined from USD 0.359 million to USD 0.228 million, while CVaR fell from USD 2.61 million to USD 2.12 million. The fifth-percentile NPV improved from USD −2.22 million to USD −1.70 million.

These comparisons show that the integrated investment strategy affected the lower tail of the portfolio return distribution as well as its expected value. At matched expected return, the integrated portfolios exhibited lower loss probabilities and smaller adverse financial outcomes.

The statistical comparisons were consistent with these findings. After Holm adjustment, the adjusted p-value was 1.7×10^{-8} for eight of the nine principal comparisons and 4.4×10^{-5} for the no-upgrade contrast. Matched-pairs rank-biserial correlations were 1.00 for the no-twin, decoupled, and mean–variance comparisons, and 0.79 for the upgrade contrast.

The calibration characteristics of the main sample are reported in Table 16.

Table 16. Model-implied characteristics of the 30 main-sample firms

Characteristic	Median	Interquartile Range	Range
Project completion probability	0.46	0.43–0.49	0.38–0.55
Per-stage success probability	0.71	0.69–0.72	0.64–0.75
Projects with positive stand-alone expected NPV	0.90	0.90–1.00	0.70–1.00
Project benefit–cost ratio	2.09	1.98–2.20	1.60–2.63
Total development-stage cost per project (USD million)	1.16	1.00–1.24	0.83–1.51
Lite-twin direct-savings payback (years)	4.4	2.9–7.2	3% without finite payback

Note: NPV = net present value.

The median project completion probability was 0.46, and the median share of projects with positive stand-alone expected NPV was 90%. The median benefit–cost ratio was 2.09, while the median payback period for the lite twin from direct operating savings was 4.4 years. These model-implied characteristics describe the investment environment represented in the simulations and provide the economic context for interpreting the portfolio comparisons.

4.8 Implications of Technology Investment Policies

The final experiment examined how public investment support affected digital twin adoption and capital allocation across 60 simulated firms. Two policy instruments were considered: capital expenditure subsidies for digital twin adoption and research and development tax credits applied to project costs.

Table 17 reports the adoption outcomes and public expenditure associated with the tested policies.

Table 17. Digital twin adoption under alternative investment support policies

Policy	Adoption Rate	Public Cost (USD Million)	Additional Adopters	Spending on Firms Adopting Without Support
No policy	62%	0	0	—
Capital subsidy 20%	68%	4.2	4	91%
Capital subsidy 40%	73%	9.1	7	84%
Capital subsidy 60%	78%	14.6	10	79%
R&D tax credit 10%	58%	19.5	−2	—
R&D tax credit 20%	55%	39.5	−4	—

Note: R&D = research and development. Additional adopters represent the difference in adoption numbers compared with the no-policy baseline. Negative values indicate that tax credits reduced the number of adopters by reallocating capital toward development projects. The em dash (—) indicates that the measure is not applicable to the respective policy.

Capital subsidies increased the adoption rate from 62% without support to 68%, 73%, and 78% at subsidy rates of 20%, 40%, and 60%, respectively. The corresponding numbers of additional adopters were four, seven, and ten firms. Public expenditure per additional adopter was approximately USD 1.05 million, USD 1.30 million, and USD 1.46 million.

A substantial share of subsidy expenditure was directed to firms that would have adopted the technology without support. This share was 91% under the 20% subsidy, 84% under the 40% subsidy, and 79% under the 60% subsidy. The results indicate that broad capital subsidies increased adoption but also financed a large number of investments that were already economically justified under the baseline conditions.

Research and development tax credits produced a different allocation effect. Adoption declined to 58% under a 10% credit and 55% under a 20% credit. Because the credit reduced the effective cost of development projects while the investment budget remained constrained, firms allocated more capital towards those projects relative to digital twin adoption. The policy therefore changed the competition for capital between enabling technology and innovation activities.

The social-value calculation added a knowledge spill-over equal to 30% of expected project revenue to private NPV before fiscal transfers. Under this specification, none of the tested policy instruments increased aggregate social value relative to the no-policy benchmark. The additional adoption induced by subsidies occurred among firms with lower private returns from the technology, while the tax credits redirected investment towards development projects.

Figure 10 shows the relationship between digital twin adoption and public expenditure, together with the share of subsidy expenditure directed towards firms that would have adopted without support.

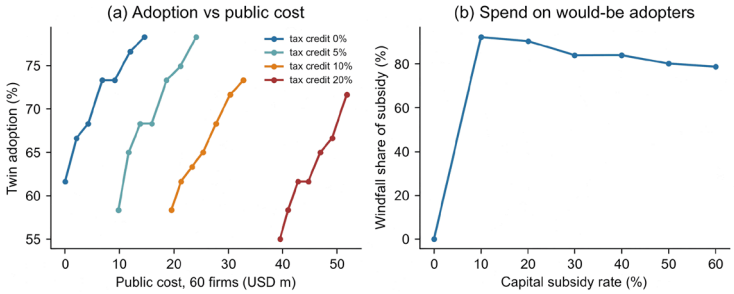


Figure 10. (a) Digital twin adoption versus public expenditure; (b) subsidy spending on firms that would adopt without support

Note: USD m = USD million.

These findings show that investment policy affects both the decision to adopt digital twins and the allocation of capital within the firm. Capital subsidies directly reduce technology acquisition costs, whereas project-level tax credits change the relative attractiveness of competing investments. The economic consequences therefore depend on the relationship between technology adoption, development portfolios, and the firm's financing constraint.

The results support the use of portfolio characteristics in evaluating technology investment policies. Firms with higher recoverable scrap values and development pipelines containing more process-related projects generated stronger economic returns from digital twin adoption. These characteristics provide a basis for distinguishing investments with substantial portfolio-level value from those justified mainly by direct operating savings.

Overall, the economic evaluation establishes that digital twin investment affects manufacturing SMEs through both operating performance and innovation portfolio outcomes. Joint optimization produced higher expected returns and more favourable downside-risk characteristics than the principal alternative appraisal procedures. The

largest contribution arose from recognising the effects of the technology on development projects, while staged investment, capital allocation, and project complementarities shaped the final adoption decision. These findings provide the basis for the technology investment and managerial implications discussed in Section 5.

5. Discussion

The results establish a distinction between the economic value of a digital twin as an individual manufacturing investment and its value as part of an innovation portfolio. Conventional appraisal primarily considers the operating savings generated by the technology, whereas portfolio-based appraisal also accounts for changes in development costs, project success, and the allocation of capital among competing activities. The TEIP framework brings these effects into a common investment decision and identifies how they influence expected returns, downside risk, and technology adoption.

The implications extend beyond the choice of an optimization method. They concern the economic unit being evaluated, the role of technological capabilities in project development, and the conditions under which staged investment and public support affect capital allocation. These issues are discussed below in relation to technology investment theory, manufacturing management, and the scope of the model.

5.1 Implications for Technology Investment Appraisal

The principal contribution of the TEIP framework is to extend digital twin valuation from an individual capital project to the development portfolio supported by the technology. A stand-alone investment calculation compares expected operating savings with acquisition and ownership costs. Such a calculation remains useful as an initial financial assessment, but it assigns no value to changes in the performance of other projects. In manufacturing firms where development activities depend on common production capabilities, these effects form part of the economic return from technology adoption.

The comparison between investment strategies illustrates the importance of this distinction. Across the main sample, integrated appraisal increased frontier hypervolume by 26.1% relative to the no-twin benchmark. The portfolio-first sequential strategy recovered most of this improvement, generating a 24.0% gain without reselecting the projects after the digital twin decision. Joint optimization added a further 1.7% relative to portfolio-first evaluation. The larger economic difference therefore arose from recognising the technology's effects on the portfolio, rather than from the additional flexibility of solving the technology and project decisions simultaneously.

This result refines the interpretation of investment interdependence. Joint optimization provides a consistent way to account for the reciprocal relationship between digital twin adoption and project selection, but the main valuation error occurs earlier, when the technology is assessed independently of the activities it supports. For firms with an established development portfolio, evaluating the digital twin against that portfolio captures a substantial share of the integrated economic gain. Full joint optimization remains valuable when adoption changes the preferred combination of projects or the use of available capital.

The channel-ablation results explain why portfolio-aware appraisal changes the investment decision. Direct operating savings alone accounted for approximately 38% of the full-model hypervolume improvement, leaving 62% associated with the combined development-project channels. Changes in stage-success probabilities and development expenditure made the largest positive contributions. The technology therefore generated economic value not only by reducing production losses but also by changing the expected cost and commercial return of innovation activities.

This finding connects technology investment appraisal with the real-options interpretation of organisational capabilities. Kogut & Kulatilaka (2001) treated capabilities as sources of strategic flexibility, while the present model expresses part of that value through changes in project-level cash flows and portfolio outcomes. A digital twin can influence the economic prospects of several development activities without being treated as a separate revenue-generating project. Its value is consequently related to the composition of the development pipeline and the opportunities created by its use.

The portfolio results also qualify the conventional interpretation of diversification. Lower dependence between project failures is generally associated with reduced exposure to common adverse outcomes. In a development portfolio containing complementary projects, however, successful outcomes may also occur together. Reducing dependence can lower the probability that complementary projects complete successfully in the same investment scenario, thereby reducing expected synergy returns.

The analytical result in Proposition 2 establishes this relationship for the one-factor Gaussian model with fixed marginal stage-success probabilities and non-negative complementary returns. The numerical experiments showed that the reduction in expected synergy value outweighed the corresponding improvement in downside risk for the simulated portfolios. Technology appraisal must therefore consider the economic consequences of changing dependence, rather than treating every reduction in outcome correlation as an increase in portfolio value.

A further contribution concerns the relationship between direct investment thresholds and portfolio-level

economic value. Proposition 1 provides a closed-form break-even condition based on recoverable scrap value and the present value of technology ownership costs. For the reference lite twin, the stand-alone threshold was approximately USD 0.384 million per year. The corresponding portfolio-aware threshold was approximately USD 0.17 million per year, reflecting the additional economic contribution of the development portfolio.

The difference between these thresholds has a practical interpretation. Under the expected-value conditions of Corollary 1, it corresponds to an externality-equivalent present value of approximately USD 0.23 million. This converts the portfolio contribution into monetary units that can be compared with acquisition costs and alternative investment opportunities. The portfolio-aware threshold also depends on the project pipeline, development-stage characteristics, and capital allocation conditions, making it a firm-specific investment measure.

The population analysis reinforced this distinction. Stand-alone and portfolio-aware appraisal produced different adoption decisions in 24% of the 400 simulated firms. Most disagreements arose when stand-alone appraisal rejected a digital twin that generated sufficient value through development projects to justify adoption. The reverse also occurred when the technology's capital requirement displaced more valuable project investments. The economic contribution of a digital twin is therefore determined by both the activities it supports and the opportunities forgone when capital is committed to the technology.

Together, these findings establish a portfolio-based interpretation of manufacturing technology investment. The digital twin is evaluated as a technological capability whose economic effects extend across development projects, while real-options reasoning accounts for the timing and flexibility of the capital commitment. Combining these perspectives provides a more complete basis for investment appraisal under capital and downside-risk constraints.

5.2 Managerial and Policy Implications

For manufacturing SMEs, the results provide a practical sequence for evaluating digital twin investment. The first step is to establish the direct economic case using recoverable scrap value, ownership costs, expected operating effectiveness, and the useful life of the technology. The stand-alone break-even condition in Eq. (9) provides this initial assessment without requiring portfolio optimization.

Recoverable scrap value can be estimated from annual production value and the proportion of production losses that the technology is expected to address. The calculation should use the firm's own operating records and the anticipated scope of the digital twin implementation. Acquisition costs, integration expenditure, annual operating costs, and the discount rate determine the corresponding investment threshold. This approach places the initial decision on measurable production and financial quantities.

The second step is to evaluate the technology against the firm's development portfolio. Where the direct-savings calculation does not justify adoption, the digital twin may still generate sufficient economic value by improving project success or reducing development expenditure. The firm should therefore identify the projects that can use the technology, their remaining development stages, expected costs, commercial returns, and relationships with other projects.

The TEIP model provides a procedure for translating this information into a portfolio-aware investment decision. Portfolios can be evaluated with and without the digital twin under the same budget and risk preferences. The difference in expected economic performance identifies the contribution associated with adoption, while the resulting project selections show how the technology changes the allocation of capital. This comparison is particularly relevant for firms with development pipelines containing a substantial proportion of process-related projects.

The population analysis showed that process-project share affected portfolio-aware adoption more strongly than stand-alone appraisal. Across quartiles of process-project share, portfolio-aware adoption increased from 65% to 77%, while the mean utility gain rose from USD 0.42 million to USD 0.78 million. These results indicate that the composition of the development pipeline should be considered alongside production losses when assessing the economic case for a digital twin.

The third step concerns the scale and timing of implementation. The staged investment results support an initial lite deployment followed by an upgrade decision informed by observed technology efficacy. The economic value of this arrangement arises from the ability to make the larger investment commitment after obtaining information about operating and development-project effects.

At the baseline efficacy variation and signal-noise settings, the estimated flexibility value was USD 0.056 million. Under higher efficacy variation, the value increased to USD 0.273 million, slightly exceeding the initial capital cost of the lite system. The results establish the economic importance of structuring early deployment around information that is relevant to the subsequent investment decision.

A staged implementation should therefore include performance measures capable of informing the upgrade choice. In addition to scrap reduction and production efficiency, these measures should cover development-stage completion, qualification expenditure, and the use of the digital twin across funded projects. The value of the initial implementation depends partly on the information it provides about the economic return from expanding the technology.

Capital allocation remains central throughout this process. The digital twin competes with development projects for investment resources, and its adoption can change which projects should receive funding. The integrated model addresses this trade-off by evaluating technology deployment and project selection under a common budget. The resulting investment decision reflects both the direct cost of adoption and the expected economic consequences for the development portfolio.

The policy experiments extend this capital allocation perspective to public investment support. Capital expenditure subsidies increased digital twin adoption, but a substantial proportion of public expenditure was allocated to firms that would have adopted without assistance. Under the tested subsidy rates, this proportion ranged from 79% to 91%. Broad subsidies therefore increased adoption while directing much of the available support towards investments already justified by private economic returns.

Research and development tax credits produced a different result. By reducing the effective cost of development projects, the credits increased their relative attractiveness under the firm's investment budget. Digital twin adoption consequently declined in the simulated policy population. The result illustrates how support for one category of innovation expenditure can alter investment in complementary technological infrastructure.

These findings connect with the broader literature on the relationship between public support and private research investment. David et al. (2000) examined the complementarity and substitution effects associated with public research and development support. In the present model, the allocation effect arises from the competition between technology adoption and development projects within the firm's capital constraint. Policy instruments that alter relative investment costs consequently influence both the adoption decision and the composition of the innovation portfolio.

The results provide an economic basis for considering more targeted forms of technology support. Recoverable scrap value, development-project composition, and technology ownership costs can be used to identify firms for which digital twin adoption generates substantial portfolio-level returns. Support linked to staged deployment or the financing of complementary development activities can also address the relationship between technology acquisition and project investment.

For firms and policymakers alike, the central implication is that digital twin adoption should be evaluated in relation to the manufacturing and development activities it supports. The economic return from the technology depends on more than the operating savings recorded after installation; it also reflects changes in development performance, investment timing, and the allocation of financial resources.

5.3 Limitations and Future Research

The TEIP framework was evaluated using simulated firms generated from parameter ranges selected to represent additive manufacturing SMEs. This design allowed the investment mechanisms to be examined systematically across different production conditions, project characteristics, and technology costs. The reported adoption rates, portfolio gains, and investment thresholds describe the behaviour of the model under these settings.

The baseline analysis represented firms with ten candidate development projects and a digital twin offering lite and full implementation levels. The robustness experiments extended the project set to fifteen, while the hold-out analysis considered firms with eight to fourteen projects and wider ranges of operating and financial characteristics. These variations established the performance of the investment framework across the tested configurations.

The model represented project dependence through a common Gaussian factor and allowed digital twin efficacy to influence operating savings, development costs, stage-success probabilities, and correlation. Project-specific efficacy experiments introduced additional variation across development activities, while alternative spill-over settings examined the economic effects of concentrating the technology's influence on process-related projects. These analyses identified how the distribution of technological effects across projects influenced investment value.

The investment horizon and decision structure were also specified in advance. The baseline model used a staged upgrade decision at the end of year 1 and did not include subsequent rounds of portfolio reselection. Extending the framework to repeated investment decisions would allow project progress, operating information, and changes in financial resources to influence technology and portfolio choices over multiple periods.

The economic comparisons used expected NPV and CVaR to represent return and downside exposure. Hypervolume provided a measure of multi-objective frontier performance, while expected NPV at matched CVaR supplied a monetary comparison independent of the hypervolume reference point. The principal optimization experiments used NSGA-II, whereas the population and policy analyses employed hill-climbing under risk-adjusted utility objectives. These procedures served different experimental purposes and were interpreted within their respective evaluation settings.

The analytical investment thresholds were derived from simplified cash-flow relationships. The stand-alone threshold captured direct savings and ownership costs, while the portfolio-aware threshold incorporated the economic contribution of development projects. Their application to an individual firm requires its own production data, development pipeline, technology costs, and investment preferences.

Further work can extend the framework through calibration with observed manufacturing and project-level data.

Records of production defects, stage-gate outcomes, development expenditure, and digital twin performance would permit the model parameters to be estimated for specific firms. Repeated observations could also support dynamic portfolio decisions in which technology capabilities and project characteristics change over time.

Additional applications can examine firms operating under different manufacturing processes, technology acquisition arrangements, and financing conditions. Extending the model to inter-firm competition and shared technological infrastructure would also allow the economic effects of digital twin investment to be studied beyond a single enterprise. At the policy level, firm-level calibration would support comparisons of targeted subsidies, staged investment assistance, and programmes addressing the joint financing of enabling technologies and development projects.

The present study establishes the portfolio valuation framework and its economic mechanisms through simulation, analytical relationships, and independent evaluation experiments. These results provide a foundation for applying the approach to manufacturing investment decisions using firm-specific operating and development data.

6. Conclusions

This study developed an integrated economic evaluation framework for digital twin investment in additive manufacturing SMEs by combining staged real options with multi-objective innovation portfolio optimization. The TEIP model accounted for direct operating savings, development costs, stage-success probabilities, project interdependencies, and downside risk under capital constraints.

Across 30 simulated firms, the integrated strategy increased risk–return frontier hypervolume by 26.1% relative to portfolios without digital twins and by 6.2% relative to stand-alone technology appraisal. Evaluating the digital twin against an optimized portfolio accounted for most of the improvement, while joint portfolio reselection contributed a further 1.7%. The channel-ablation analysis associated approximately 62% of the modelled improvement with effects on development projects rather than direct operating savings.

The analytical results established a stand-alone investment threshold based on recoverable scrap value and identified the additional economic contribution incorporated in portfolio-aware appraisal. Across 400 simulated firms, the two appraisal procedures produced different adoption decisions in 24% of cases. The staged upgrade analysis also showed that information about technology efficacy contributed economic value by guiding the timing of further investment. Confirmation, hold-out, and downside-risk evaluations supported the principal portfolio results across the tested conditions.

The findings indicate that digital twin investment should be assessed in relation to the development activities it supports and the capital allocation decisions it changes. The proposed framework connects technology-level economic appraisal with portfolio selection and staged investment, providing a basis for evaluating adoption thresholds, investment sequencing, and downside exposure. For manufacturing SMEs, this approach supports capital allocation between digital twin implementation and competing innovation projects. For policymakers, it provides a framework for examining how technology support measures affect adoption and the financing of complementary development activities.

Data Availability

The code used in this study, including the model, NSGA-II optimization, benchmark comparisons, sensitivity analysis, break-even maps, policy analysis, population study, learning analysis, robustness tests and figure generation, is available from the corresponding author upon reasonable request.

Conflicts of Interest

The author declares no conflicts of interest.

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Appendix

Table A1. Main parameters

Parameter	Value or Range	Basis
Projects per firm; stages per project	10; 2 to 4	Design choice; 15 projects in Table 5, 8 to 14 in hold-out
Stage cost c_i; base success p_i	U (0.15, 0.6) USD million; U (0.55, 0.85)	Assumption; stage-gate structure after Cooper (1990); implied completion probability in Table 16
Annual revenue R_i	$c_i \cdot n_i \cdot$ U (0.55, 1.7), for 5 years	Assumption; scaled to a benefit–cost ratio near 2 (Table 16)
Synergy pairs; strength	8 pairs; U (0.1, 0.3) of min (R_i, R_j)	Assumption; synergy mass tested as a driver in Section 4.5
Systematic correlation $\rho 0$	U (0.2, 0.5)	Assumption; one-factor form after Vasicek (2002); ρ_0 = 0.6 in Table 5
Defect rate q; production value V	U (3%, 15%); U (3, 10) USD million/year	Assumed SME range; widened in the hold-out firms
Efficacy θ	lognormal, mean 1, CV U (0.3, 0.8)	Assumption; mean 1 by normalisation (levels set in Table 2); CV to 1.0 in Table 5 and 1.8 in Table 12
Learning signal noise	0.35 (log scale)	Assumption; varied from 0.05 to 3.0 (Table 12)
Discount rate; twin horizon	10%; 8 years	Conventional rate; 8 to 12% and 5 to 10 years in Table 8
CVaR level; budget	5%; 50% of full-portfolio expected outlay	Convention; 1% and 10% and budgets of 30% and 70% in Table 5
Scenarios: optimization; validation	3,000; 20,000	Numerical choice; convergence check in Section 4.1
Hold-out ranges (30 firms)	q U (1%, 20%); V U (1.5, 14) USD million/year; $\rho 0$ U (0.1, 0.7); CV U (0.2, 1.2); cost scale U (0.7, 1.8); 8 to 14 projects; budget 30%, 50% or 70%	Deliberately wider than the main ranges
Project-specific efficacy; spill-over weight	σ_i in {0, 0.3, 0.6, 1.0}; weight in {0, 0.3, 0.6}	Robustness settings (Table 13)
Population study; learning study	400 firms, 1,500 scenarios, U = E − 0.5·CVaR; 10 firms, CV 0.1 to 1.8, σ_s 0.05 to 3.0	Separate experiments (Sections 4.5 and 4.6)

Table A2. Notation

Symbol	Meaning
N, n_i	Number of candidate projects; number of remaining stage gates of project i
c_i, p_i, R_i	Stage cost, base per-stage success probability and annual revenue of project i
φ_i, w_i	Process-related flag and effective twin exposure of project i
s_{ij}	Synergy between projects i and j
Z, ρ, ρ_0	Systematic factor; factor loading after the twin; base factor loading
θ, S	True twin efficacy; noisy efficacy signal observed after year 1
η_ℓ, β_ℓ, γ_ℓ, $\kappa_{d,\ell}$	Twin effects on savings, success logit, stage cost and correlation at level ℓ
x, d_0, κ	Project selection, lite-twin deployment and upgrade threshold
r, α, B	Discount rate, CVaR tail level and expected-outlay budget
V, q	Annual production value and defect rate (recoverable scrap value is $V \cdot q$)
TEI, ΔU	Twin Externality Index, defined as $\Delta U - \max(E_{sa}, 0)$
Λ	Flexibility value of the staged upgrade option.